

CHAMP and SAC-C atmospheric occultation results and intercomparisons

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[1] The German Challenging Minisatellite Payload (CHAMP) and Argentine Satellite de Aplicaciones Cientificas-C (SAC-C) Earth science missions, launched in 2000, carry a new generation of Global Positioning System (GPS) receivers for radio occultation sounding of the ionosphere and neutral atmosphere. Though the occultation concept for obtaining profiles of atmospheric temperature, pressure, and moisture was proven in 1995 with GPS/MET, concurrent measurements from CHAMP and SAC-C present the first opportunity for a preliminary evaluation of three central claims: (1) GPS soundings are effectively free of instrumental bias and drift; (2) individual temperature profiles are accurate to <0.5 K between ~ 5 and 20 km; and (3) averaged profiles for climate studies can be accurate to <0.1 K. These properties imply that a weak climate trend can be monitored and detected in less than a decade and studied by different instruments at different times with no external calibration. While this detection cannot by itself tell us the source of the climate change, whether natural and anthropogenic, this detection is a prerequisite to answer the more difficult problem of understanding the cause of change. In this paper, these three claims are evaluated by comparing nearby CHAMP and SAC-C profiles. Of nearly 130,000 profiles examined, 212 pairs occurring within 30 min and 200 km of one another were found. Profile pairs agree to <0.86 K (68% confidence interval) and to within 0.1 K in the mean between 5 and 15 km altitude, after removing the expected variability of the atmosphere. If the errors in CHAMP and SAC-C are assumed to be uncorrelated, this implies that individual profiles are precise to <0.6 K between 5 and 15 km. Individual comparisons show closest agreement near the tropopause and display finer resolution than and substantially different temperatures from numerical weather model analyses from the European Centre for Medium-Range Weather Forecasts (ECMWF). Comparisons between CHAMP and SAC-C largely indicate precision; however, several features observed in common, especially near the tropopause, tend also to indicate accuracy. Limitations of previous experiments (e.g., GPS/MET) in probing the lower troposphere have significantly improved with CHAMP and SAC-C, with the majority of profiles (60%) descending to the lowest 0.5 km. This is expected to increase to 90–95% with future system improvements. However, the N-bias problem encountered in GPS/MET is also present in CHAMP and SAC-C, and it is expected to be much reduced once open loop tracking is implemented. Examples are selected to illustrate lower tropospheric sensing, including detection of the planetary boundary layer height. For the first time, such performance is achieved with GPS Antispoofing encryption on. Daily occultations currently number ~ 350 –400; this is expected to reach over 1000 in the near future, rivaling the number of semidaily radiosonde launches. With several new missions in planning, this may increase tenfold in the next 3–8 years, making GPS sounding a potentially significant input to numerical weather prediction and climate research. *INDEX TERMS:* 0350 Atmospheric Composition and Structure: Pressure, density, and

temperature; 1640 Global Change: Remote sensing; 0394 Atmospheric Composition and Structure: Instruments and techniques; 6904 Radio Science: Atmospheric propagation; 1241 Geodesy and Gravity: Satellite orbits; **KEYWORDS:** GPS occultations, temperature profiling, CHAMP, SAC-C

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1. Introduction

[2] The concept of atmospheric profiling by GPS occultations was first introduced in 1988 [Yunck *et al.*, 1988] and later demonstrated with the GPS/MET experiment in 1995 [Ware *et al.*, 1996]. The concept is derived from planetary occultation experiments where measurements of signal time delay of a spacecraft occulted behind a planet as viewed from Earth is used to infer properties of the planet's atmosphere [e.g., Fjeldbo *et al.*, 1971]. Applied to Earth, the transmitter is a GPS satellite and the receiver is placed on a low-Earth orbit (LEO). The fundamental measurement is the time delay of the transmitted GPS signal as the satellite sets or rises behind Earth's atmosphere. Precise time delay measurements are converted into atmospheric Doppler shift and bending, from which atmospheric refractivity, density, temperature, pressure and water vapor can be inferred.

[3] After the successful demonstration of this concept with the GPS/MET experiment in 1995 [e.g., Ware *et al.*, 1996; Hajj *et al.*, 1996; Kursinski *et al.*, 1996; Leroy, 1997; Rocken *et al.*, 1997; Kursinski and Hajj, 2001], several challenges remained: (1) obtaining accurate retrievals when the GPS signal is encrypted (referred to as Anti-spoofing or AS) by the Department of Defense (DoD) (AS on is the normal mode of operation for GPS; during the GPS/MET experiments in 1995–1997, AS was turned off for 4 three-week periods for the sake of demonstrating GPS profiling); (2) obtaining a sufficient density of occultations to facilitate scientific and operational use of the data; (3) penetrating the lower troposphere in moist regions; (4) demonstrating sub-Kelvin temperature accuracy; and (5) understanding and possibly eliminating causes of retrieved refractivity biases in the lower troposphere.

[4] Overcoming these challenges is crucial to meeting the full potential of GPS occultation science; applications include improving our understanding of the troposphere-stratosphere exchange, water vapor circulation, heat and energy transport by gravity waves, climate monitoring and fingerprinting, and weather prediction. GPS occultations promise to provide global atmospheric profiling with high vertical resolution (0.1–1 km) and high accuracy (<1 K temperature, <0.5 g/kg specific humidity) under all weather conditions. Moreover, because GPS occultation is an active measurement, it yields an independent estimate of pressure as a function height, therefore, enabling the determination of geopotential heights with an accuracy reaching 1–2 m. The geopotential height of a pressure level is a measure of the atmospheric weight above it and can be used as a sensitive gauge of climate trends.

[5] The launch of the German Challenging Minisatellite Payload (CHAMP) and the Argentine Satellite de Aplicaciones Científicas-C (SAC-C) missions in 2000, carrying a

new generation of GPS receivers, enables us to address the challenges listed above. In this paper we discuss the first four challenges in conjunction with the current missions. The fifth challenge, where retrieved refractivity is systematically smaller than those of numerical weather prediction models (also known as the negative-N bias problem) has been addressed elsewhere [Sokolovskiy, 2001a, 2003; Beyerle *et al.*, 2003; Ao *et al.*, 2003] and will only be mentioned briefly in this paper. The first challenge is partially solved by use of a new generation “semicodeless” GPS receiver, which enables accurate tracking of the GPS signals while AS is turned on. The second challenge is met as both missions continuously provide 350–400 occultations every day, a number that will soon reach 1000+ once rising occultation capability is implemented on board the SAC-C satellite and occultations from other flying missions (e.g., GRACE) are turned on. (Because of numerous software uploads and tests on SAC-C, including the implementation of open loop tracking, coverage from this satellite has varied considerably over the past two years.)

[6] In comparison with the earlier GPS/MET instrument, these new receivers substantially improve our ability to sense the lower troposphere. Here we present results demonstrating consistent penetration down to the lowest kilometer of the atmosphere for most of the CHAMP and SAC-C occultations. With routine sensing of the lower troposphere, GPS occultations provide the first space-based remote sensing technique that is potentially able to sense the PBL and the transition into the free troposphere with a high vertical resolution.

[7] A substantial part of this paper is devoted to addressing the fourth challenge, to demonstrate sub-Kelvin accuracy. This challenge can be broken into three central claims: (a) GPS occultation profiles are free of instrumental drift and are virtually unbiased. (b) Individual GPS temperature retrievals have 0.5 K accuracy between ~5 and 20 km and approach 0.2 K accuracy near the tropopause. (c) Averaged profiles for use in climate studies can reach a temperature accuracy of 0.1 K or better at these altitudes.

[8] If these claims can be verified, they would establish GPS occultation as the most accurate spaceborne technique for measuring tropospheric and stratospheric temperatures and for monitoring subtle climate variabilities.

[9] The difficulty in testing these claims stems from the lack of current measurements or models more accurate than about 1 K, making comparisons with other data sources of limited value. Such comparisons have been performed by numerous groups (see, e.g., Kursinski *et al.* [1996] or Rocken *et al.* [1997] for GPS/MET; Wickert *et al.* [2001] or Hajj *et al.* [2002a] for CHAMP) and have established that GPS temperatures are consistent with those of radiosondes, satellite radiometers and weather forecasts to about 1–2 K between 5 and 25 km. While such comparisons are

valuable, they cannot tell us which of the compared quantities is more accurate. And while detailed theoretical analysis [e.g., *Kursinski et al.*, 1997] suggests GPS soundings to be accurate to better than 0.5 K between 5 and 25 K and about 0.2 K at the tropopause, this has not yet been demonstrated. The presence now of two sensors taking GPS measurements independently from different vantage points provides our first opportunity to test such claims and to establish better precision estimates for current GPS retrievals. The difficulty in establishing accuracy by use of CHAMP and SAC-C intercomparison is that they both can be susceptible to similar systematic errors. Those include residual ionospheric errors, negative N-biases, dependence on the retrieval scheme, dependence on the initialization of the Abel inversion and the hydrostatic equilibrium integral, and horizontal averaging. Therefore our emphasis in this paper is more on precision than on accuracy; however, many features that are commonly observed by CHAMP and SAC-C, especially near the tropopause, also indicate accuracy.

[10] We should stress that claims on temperature precision are valid only in regions where effects of water vapor on the occulted signal can be ignored. Since the saturation water vapor pressure decreases rapidly with decreasing temperature, as dictated by the Clausius-Clapeyron equation, water vapor in tropospheric regions where $T < 250$ K can be ignored [*Kursinski et al.*, 1995], and sub-Kelvin accuracy in these regions is expected. This condition extends down to the surface near the pole, making this technique promising for monitoring climate variability at the Arctic and Antarctic regions.

[11] In regions where water vapor becomes significant, accuracy claims can be stated in terms of refractivity (N). Early simulations showed that N should be accurate to $\sim 0.2\%$ above 10 km growing to 1% near the surface because of horizontal gradient errors [*Kursinski et al.*, 1997]. Early validation of refractivity against weather analyses has shown a negative bias in the lower troposphere [*Rocken et al.*, 1997]. The same bias is also seen in the CHAMP and SAC-C data [*Marquardt et al.*, 2003; *Ao et al.*, 2003] and its causes are discussed briefly later. In order to avoid complications arising from the N-bias, we limit our validation efforts to temperatures derived in dry regions, where $T < 250$ K or in the stratosphere.

[12] Occultations from CHAMP and SAC-C have been collected since the middle of 2001. By the middle of June 2003, the two missions have gathered $\sim 450,000$ occultations. To keep up with the stream of occultation data, a versatile data analysis system is set up at the Jet Propulsion Laboratory (JPL), allowing us to process and examine the data continuously. Our presentation starts with an overview of this occultation processing system (section 2).

[13] A prerequisite to obtaining high-quality retrievals is the computation of precise orbits of the GPS and the LEO satellites. Accuracies of CHAMP and SAC-C orbits are examined in section 3. In sections 4–7 we address, in order, challenges 1–4 listed above. In section 4 we examine the signal noise properties and how they map into retrieved temperature errors. We also perform statistical comparisons to weather analyses to assess current retrieval accuracy with AS on compared to that of GPS/MET with AS off. In section 5, we examine the number of daily occultations

collected by both missions and their coverage. Section 6 addresses current performance in sensing the lower troposphere. This is done by showing specific examples and by statistical analysis to assess the degree of success in penetrating the lower troposphere. In section 7 we directly compare CHAMP and SAC-C occultations, presenting individual examples and statistics. Conclusions are summarized in section 8.

2. An Overview of the JPL Occultation Processing System

[14] The occultation data analysis system at JPL consists of a number of processes that can be grouped under five main categories (Figure 1):

[15] 1. The first is data collection: In this process, flight and ground data are collected, reformatted, corrected for bit errors and flagged for cycle slips. Ground data consist of two types, medium-rate data collected at 1 Hz, and low-rate data collected at 1/30 Hz. The former is used for GPS-occultation calibration in which the GPS clocks are solved for at the 1-s rate and removed. (The demise of GPS clock dithering, known as selective availability, has made it possible to use low-rate ground data for GPS-occultation calibration; see *Wickert et al.* [2002]). The latter is used to solve for the GPS orbits.

[16] 2. The second is orbit determination: Both GPS and LEO orbits are needed for the processing of occultation data. Three types of GPS daily orbit solutions are routinely generated at JPL with different production speed and accuracy: (1) The “FLINN” final orbits, produced about 1 week after the fact, with an orbit accuracy of ~ 5 cm; (2) “QuickLook” orbits, produced daily, with an accuracy of ~ 15 cm; and (3) continuous real-time orbits accurate to ~ 0.25 m. Our current occultation processing adopts the QuickLook orbits, allowing us to produce retrievals within 24 hours of data collection. For use in numerical weather prediction, where < 3 hours of delay is required, one could adopt the real-time GPS orbits with little performance penalty. In that case the latency will depend largely on the downlink time of the LEO data.

[17] 3. The third is the calibration process: In this process, the atmospheric delay induced on the occulted link is isolated by removing all clock drifts and the geometric delay change due to the motion of the satellites. Our normal processing system uses four links, LEO-GPS1, LEO-GPS2, Ground-GPS1 and Ground-GPS2 in a double-differencing geometry to solve for and remove all clock errors. The technique is widely used in geodesy. Its application to GPS occultation is described in detail by *Hajj et al.* [2002b]. More recently, it has been shown that with SA turned off, a single difference (LEO-GPS1 minus LEO-GPS2) is sufficient for accurate temperature retrievals [*Wickert et al.*, 2002]. Since further assessment of single differencing is still needed to establish its true performance, our routine analysis still employs double differencing.

[18] 4. The fourth is the retrieval process: At this point, the atmosphere-induced Doppler and bending are derived by use of the geometric optics approximation, the ionosphere is removed by taking a linear combination of the L1 and L2 bending at the same impact parameter and the Abel inversion is performed on the calibrated bending to derive

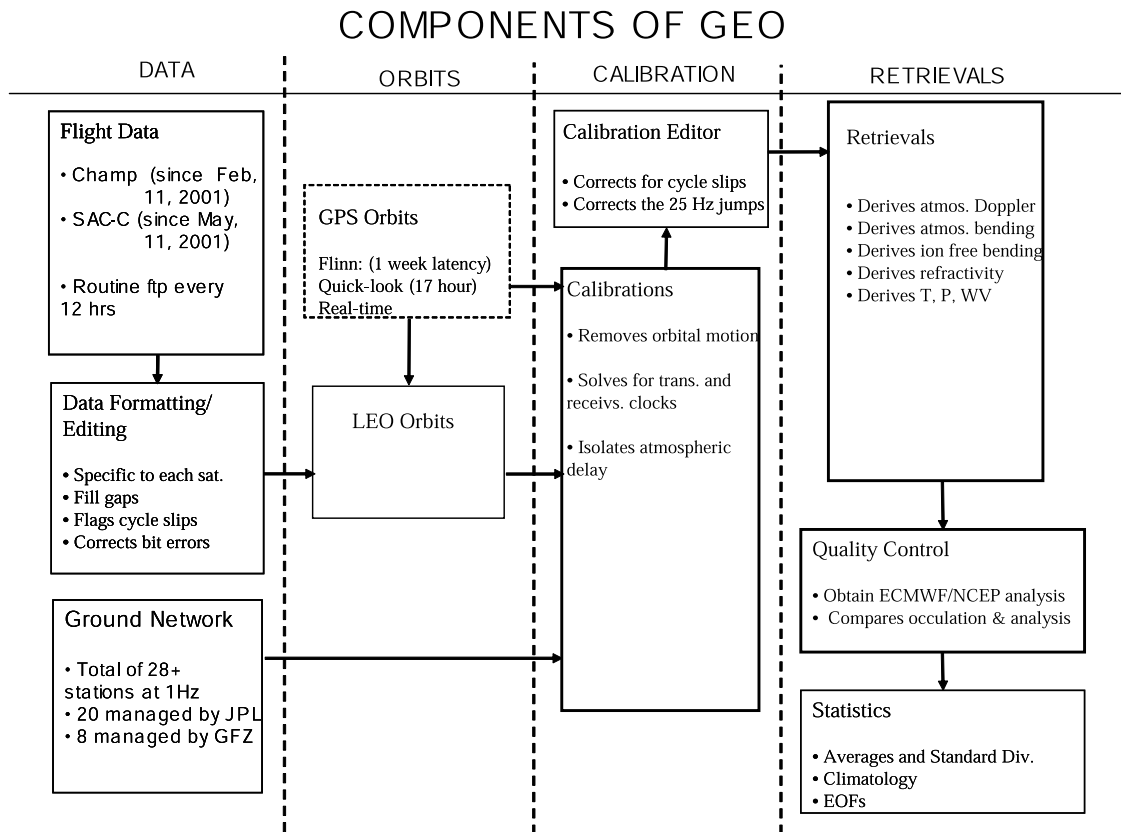


Figure 1. An overview of the GPS Earth Observatory (GEO) system at JPL used for routine processing of GPS occultation data.

refractivity. Once refractivity is obtained, the hydrostatic integral is initialized by using the temperature at a specified altitude obtained from a weather analysis. When the National Centers for Environmental Prediction (NCEP) final analysis is used, this altitude is about 30 km (corresponding to the highest pressure level of the analysis). When the ECMWF analysis is used, the hydrostatic is initialized at 40 km altitude. Similarly because of the “moist-dry” ambiguity in interpreting the refractivity in the lower troposphere, the analysis temperature is used in the lower troposphere at levels where $T > 250$ K in order to derive water vapor pressure from the GPS occultations. The detail of the processing is described by Hajj *et al.* [2002b].

[19] 5. The fifth is the quality control process: An occultation is considered successful only if its absolute temperature difference from the analysis is less than 10 K and its absolute fractional refractivity difference from the analysis is less than 20% everywhere below 30 km. Statistical comparison of GPS occultations to ECMWF and NCEP analyses show that temperatures agree to <2 K (1-sigma) and fractional refractivities agree to $<4\%$ (1-sigma) between 0 and 30 km [Kursinski *et al.*, 1996; Rocken *et al.*, 1997; Wickert *et al.*, 2001; Hajj *et al.*, 2002a]. Therefore the 10 K and 20% refractivity windows, chosen for quality control, correspond to more than 5-sigmata of the established differences, therefore constituting a very loose criterion. However, one should keep in mind that if the analysis is systematically biased in certain regions, this criterion can result in selecting bad retrievals or rejecting good ones. Therefore the type of quality control used should

be considered in conjunction with the type of investigation one seeks to address.

[20] A key element of the occultation analysis system is an automated executive system that triggers each process upon completion of the previous one. Once the data are downlinked from the satellite and transferred to JPL, the executive system fetches the required GPS orbits and triggers the LEO-orbit process. Completion of the LEO orbit then triggers calibration followed by retrieval and quality control. As discussed in more detail later, each occultation can fail any one of these processes for different reasons. (The percentage of occultations failing the different processing step can be read from Figure 9, where the NCEP analysis is used). These processes can run on different machines and for different occultations simultaneously allowing maximum flexibility, speed, and autonomy.

3. CHAMP and SAC-C Orbits

[21] Since the fundamental measurement during an occultation is the time delay between the transmitted and received signal, obtaining accurate orbits is necessary to remove orbital motion effects and isolate the atmospheric delay. The precision and accuracy of the orbits also indicate the health of the receiver in general and the quality of the data used in doing precise orbit determination (POD). In this section we discuss the precision and accuracy of our orbital solutions for CHAMP and SAC-C.

[22] The orbits of both CHAMP and SAC-C are determined using the GPS measurements collected by a zenith-

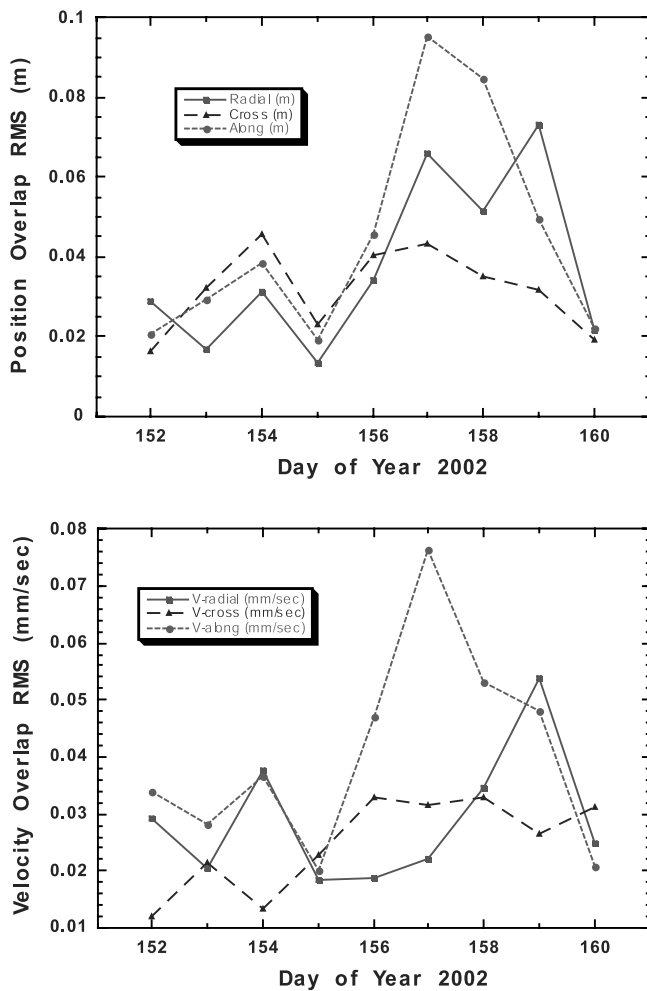


Figure 2. Daily RMS of radial, cross-track, and along-track orbital overlap differences for CHAMP (top) position and (bottom) velocity.

looking antenna. Both the zenith-looking and the aft-looking (used for occultations) antennas are connected to the same GPS receiver. Because of the low altitudes of the SAC-C (710 km) and CHAMP (410 km) satellites operating close to solar-maximum conditions, their motions are subject to sizable geopotential and atmospheric drag perturbations. The orbit determination task would require precise modeling of these perturbations if it relied on a purely dynamical approach. At JPL we apply a reduced-dynamic strategy in which local stochastic accelerations are adjusted to compensate for mismodeled forces. By tightening or loosening the constraint on the stochastic accelerations, we can tune the reduced-dynamic orbit solution to optimize between dynamic and kinematic solutions [Yunck *et al.*, 1990; Wu *et al.*, 1991; Bertiger *et al.*, 1994]. In order to obtain an optimal combination of the geometric information in the measurements and the knowledge in the orbit dynamic models, different constraints were selected for CHAMP and SAC-C through separate tuning processes similar to those discussed by Kuang *et al.* [2001].

[23] In solving for CHAMP and SAC-C orbits, the GPS satellite orbits and their clock solutions (produced daily at

JPL on the basis of data from the International GPS Service ground receiver network) are used. For each day, we process 27 hours of flight tracking data (24 hours + 3 hours from the previous day) and take the last 26-hour arc as the precise orbit solution. Orbit precision is evaluated using the differences in the 2 hours of overlapping period between two consecutive 26-hour orbit solutions. The orbit overlap difference is a self-consistency test, measuring orbit precision not accuracy. However, past experience [Bertiger *et al.*, 1994] has suggested that orbit overlap difference can be a good approximation of orbit accuracy if we use it with caution and use other tests to detect possible systematic errors. Figure 2 shows the overlap difference RMS for CHAMP orbit position and velocity during a 9-day period (1–9 June 2002). Figure 3 shows the orbital overlap difference RMS for SAC-C for the same period. Both CHAMP and SAC-C show a RMS overlap differences of <8 cm for position and <0.05 mm/s for velocity in the radial, cross-track and along-track directions. Of most importance to occultation measurements is the velocity along-track component, which shows a consistency of <0.08 mm/s for all days, meeting or exceeding the velocity accuracy requirements for GPS occultations at the altitudes of consideration here (<50 km) [Melbourne *et al.*, 1994; Kursinski *et al.*, 1997].

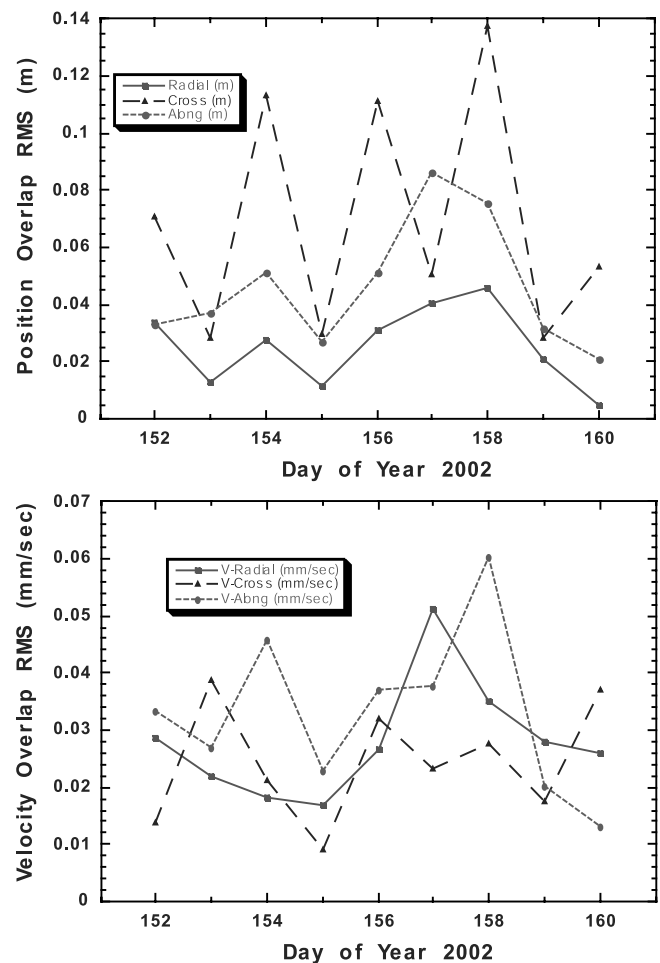


Figure 3. Same as Figure 2 but for SAC-C.

Table 1. Residual of SLR Measurements to CHAMP

Day of Year 2002	Number of Data Points	RMS, m	Standard Deviation, m
152	49	0.020	0.016
153	133	0.058	0.027
154	59	0.058	0.033
155	54	0.109	0.023
156	85	0.107	0.107
157	59	0.084	0.073
158	114	0.072	0.062
159	53	0.023	0.020
160	25	0.056	0.040

[24] A measure of the CHAMP orbit accuracy is possible by use of the laser reflector on CHAMP and Satellite Laser Ranging (SLR) data taken from a global network of ground stations. Table 1 shows the RMS and standard deviation of SLR range residuals available during the 9-day period mentioned above. Each residual is the difference between the SLR measured range and the computed range using our CHAMP orbit solution. All residuals shown here are derived from raw SLR measurements without any correction to time tag biases. The average RMS value of these SLR residuals is 7 cm, verifying that the orbit overlap difference serves as a reasonably good orbit accuracy evaluation in this case.

4. Assessment of CHAMP and SAC-C Performance

[25] A basic condition for the success of GPS/MET was the temporary disabling of the DoD anti-spoofing in which the second (L2) GPS signal is encrypted. The new generation “Blackjack” receiver on board CHAMP and SAC-C has an advanced “semicodeless” capability, which allows the recording of accurate L2 even when AS is on. In this section, we examine the fundamental observables (phase and amplitude of the occulted signal) recorded by CHAMP and SAC-C and perform statistical comparison of retrieved refractivity and temperature to the ECMWF weather analysis.

4.1. Evaluation of Basic Observables

[26] The fundamental measurements during an occultation are (a) the phase delay, which is a measure of the travel time between the occulted transmitter and receiver pair up to an unknown constant, and (b) the voltage signal-to-noise ratio (SNR) of the occulted signal, reflecting the received signal amplitude. Various techniques have been introduced in which phase [Fjeldbo *et al.*, 1971], amplitude [e.g., Sokolovskiy, 2000] or a combination of both [e.g., Melbourne *et al.*, 1994; Gorbunov *et al.*, 1996, 2000; Karayel and Hinson, 1997; Gorbunov, 2001, 2002] measurements are used to derive profiles of atmospheric refractivity. Here we examine both types of measurements for CHAMP and SAC-C.

4.1.1. Voltage Signal-to-Noise Ratio (SNR)

[27] Because of the inverse relationship between phase noise and voltage SNR ($\sigma_\phi = \text{SNR}^{-1}$ given in units of radians), an occultation with higher SNR translates into a more accurate refractivity retrieval. Table 2 shows the average 1-s voltage SNR at the start of an occultation over 1 day of CHAMP, SAC-C, and GPS/MET occultations for both L1 (~19 cm wavelength) and L2 (~24.4 cm wavelength). For GPS/MET, averages from two different periods

Table 2. Average 1-s Voltage SNR at the Start of an Occultation for Different Satellites and Modes of Operation

Satellite	Signal	Antispoofing	Starting SNR, Volts/Volt
CHAMP	L1	On	620
CHAMP	L2	On	160
SAC-C	L1	On	550
SAC-C	L2	On	100
GPS/MET	L1	On	350
GPS/MET	L2	On	35
GPS/MET	L1	Off	350
GPS/MET	L2	Off	240

are shown, corresponding to times when AS was off and on. Comparing the SNR at the beginning of the occultations, we notice that CHAMP and SAC-C have considerably higher SNR than GPS/MET. In the case of the L1 signal, this is mainly due to the 10 dB and 9 dB receiving antenna gains for CHAMP and SAC-C, versus the 6 dB maximum gain on GPS/MET (T. Meehan, JPL, personal communication, 2001). However, for L2, the substantial improvement in CHAMP and SAC-C L2 (collected with AS on), over the AS-on GPS/MET L2, is due to the advanced semicodeless tracking by the BlackJack receivers on CHAMP and SAC-C. This improvement in L2 SNR is crucial for obtaining high-accuracy retrievals with an unclassified receiver. By contrast, during AS-on periods, the low GPS/MET L2 SNR (Table 2) caused the L2 phase measurements to be noisier and, in many cases, impossible to track, precluding dual-frequency retrievals nearly 88% of the time. (The absence of accurate L2 phase measurements in GPS/MET during AS-on periods led to the development of a single frequency retrieval scheme by de la Torre Juarez *et al.* [2004]).

4.1.2. Phase Measurements

[28] An example of phase measurements from a CHAMP occultation (after removing clocks and orbital motion) is shown in Figure 4. We note that the 2.5 km extra atmospheric delay, contrasted with the 1–1.5 km delay observed

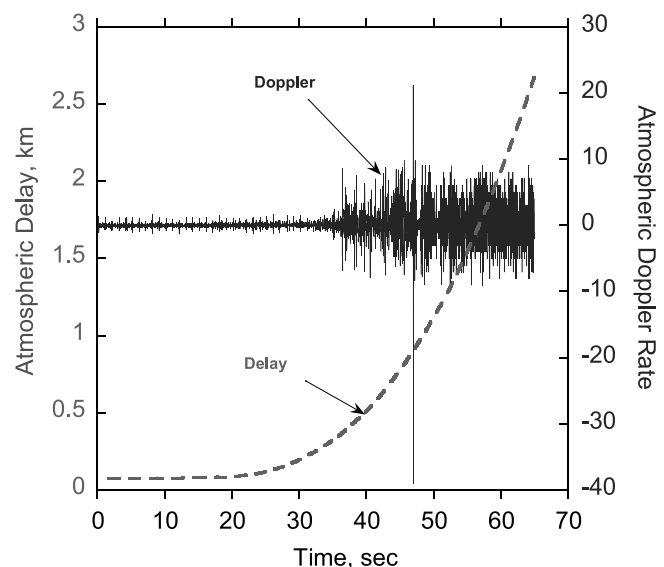


Figure 4. Phase delay (left scale) and acceleration (Hz/s, right scale) for a CHAMP occultation (with ID = 2001-10-01-13:15sacc_gps38 indicating year, month, day, UT and participating satellites, respectively).

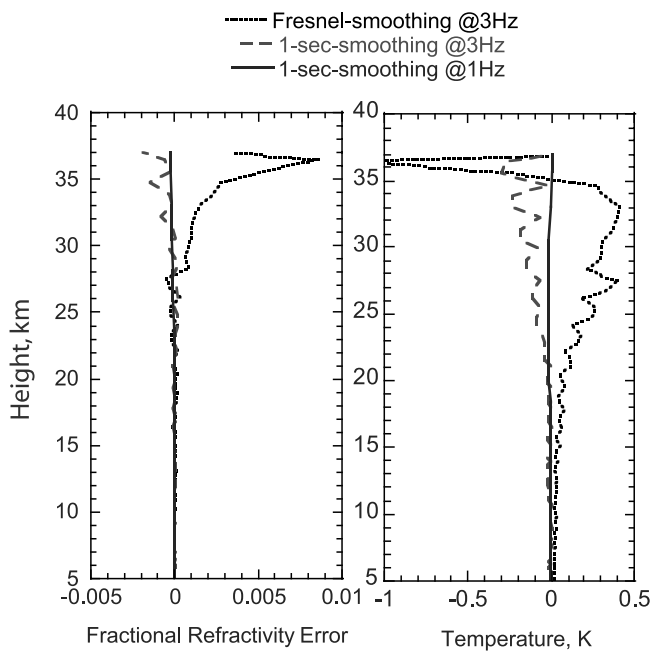


Figure 5. Fractional refractivity and temperature difference between two profiles derived from “noisy” and “smooth” phase measurements for three different smoothing windows and sampling rates.

in GPS/MET, is an indication of the deeper penetration of the occultation in the lower troposphere for CHAMP and SAC-C. We examine the noise of the phase measurements by differentiating the atmospheric delay twice with respect to time. The results are shown on the right scale of Figure 4. The significantly noisier Doppler rate toward the latter part of the occultation corresponds to when the signal is in the lower troposphere and is a result of a decreasing SNR. On the other hand, in the first half of the occultation, a most notable feature is a periodic signal at 1 Hz rate. This is due to the nonperfect cancellation of the receiver’s clock when forming the difference between the occulted and the non-occulted reference signals. This effect is more adverse for CHAMP and SAC-C than it was for GPS/MET and is due to some hardware and software implementation that are specific to CHAMP and SAC-C.

[29] Since the beginning of these two missions, numerous software uploads have been tested on the CHAMP and SAC-C receivers causing the behavior of the 1Hz signal to be quite different at different times. Therefore a detailed examination of this signal can be quite involved. However, to approximate its effect on the retrieved profile we process the occultation of Figure 4 as follows: (a) smooth the phase with a 2-s smoothing window; (b) construct a noisy phase by adding an error of +1.5 cm (corresponding to the averaged observed error) at 0.02 s before and −1.5 cm at 0.02 s after each 1-s mark of the smoothed phase; (c) retrieve the artificially constructed phase by use of three different smoothing windows and sampling times: (1) a 1-s smoothing window sampled at 1 Hz rate, (2) a 1-s smoothing window sampled at 3 Hz and (3) a variable smoothing window commensurate with the time it takes the occulted link to cross a Fresnel’s diameter size sampled at 3 Hz. (The smoothing for any given time window is done by evaluation

of a second-order polynomial fit to the phase at the middle of the window.) Figure 5 shows the difference in the retrieved refractivity and temperature between the smoothed and noisy phase measurements (obtained in steps a and b, respectively) for the three different smoothing schemes. With this procedure, and by use of the second retrieval scheme (1-s smoothing window sampled at 3 Hz), we estimate that the contribution of the 1Hz noise to temperature error is <0.2 K below 30 km and <0.1 K below 25 km. We shall see that these errors are well below the expected errors on CHAMP and SAC-C temperature profiles at these heights, and therefore, adopt the second retrieval scheme as our main approach to processing the data.

[30] Full treatments of the origin, characteristics and the effects of the 1-Hz signal are beyond the scope of this paper. The exercise given above, although simplistic, provides a sense of the effect of averaging and gives some rational to our choice of smoothing. It should be emphasized that the 1-s smoothing will also reduce atmospheric signals such as gravity waves at the corresponding wavelengths. Future GPS receivers, such as those built for COSMIC, are going through several tests to insure that this problem is either eliminated or significantly reduced such that no science information are lost.

4.2. Statistical Comparisons of Refractivity to Weather Analysis

[31] A standard method of assessing the quality of GPS occultation retrievals has been to compare them against weather analyses. Figure 6 shows means and standard deviations of fractional refractivity between profiles obtained by CHAMP and corresponding profiles obtained from 6-hourly ECMWF Tropical Ocean and Global Atmosphere (TOGA) analyses interpolated to the locations and times of the occultations. Standard deviation of fractional refractivity

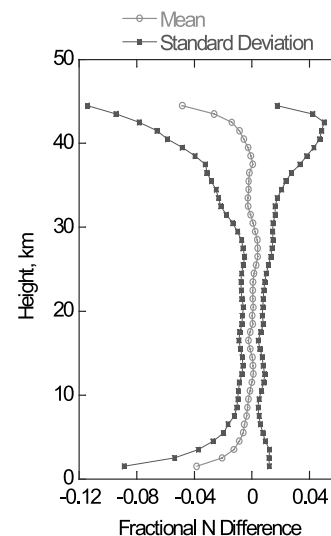


Figure 6. Mean and standard deviation around the mean of CHAMP and ECMWF fractional refractivity difference $((N_{\text{CHAMP}} - N_{\text{ECMWF}})/N_{\text{ECMWF}})$. Statistics is based on occultations collected during the period 1–7 June 2002. Corresponding statistical temperature differences are shown in Figure 7. The coverage for this period is shown in Figure 10.

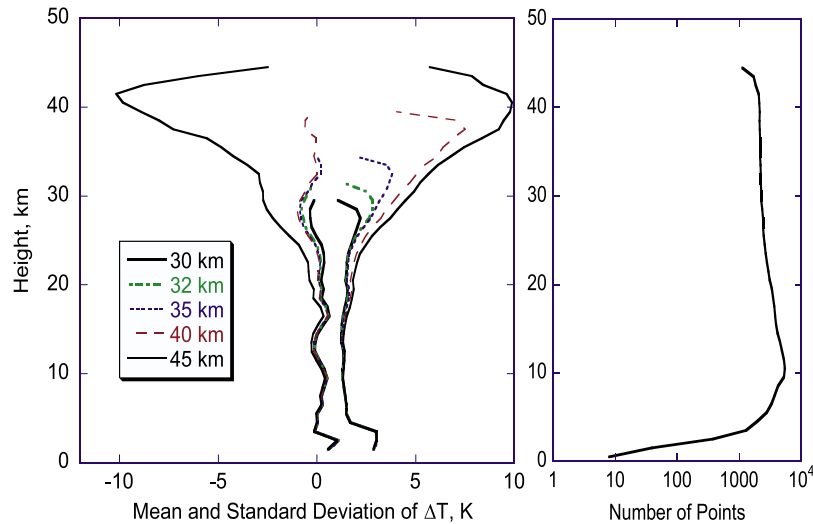


Figure 7. Mean and standard deviation of CHAMP and ECMWF temperature difference ($T_{\text{CHAMP}} - T_{\text{ECMWF}}$) with the hydrostatic integral initialized at 30, 32, 35, 40, and 45 km using the ECMWF temperature at these heights. The same set of profiles as in Figure 6 is used in the statistics. In the lower troposphere, temperature statistics are exclusive to regions where $T < 250$ K in order to avoid the dry/wet ambiguity. The number of points included in the temperature statistics at a given height is shown in the plot on the right.

differences is $\sim 1\%$ between 10 and 30 km altitudes, growing to nearly 4–6% outside this region. While there is no apparent fractional refractivity bias ($< 0.5\%$) between 10 and 30 km, significant biases are seen below and above these heights. Observed refractivity biases in the lower troposphere are mainly due to (1) receiver tracking error and (2) a half cycle ambiguity in determining the phase delay [Sokolovskiy, 2001a; Beyerle et al., 2003; Ao et al., 2003]. Open-loop tracking will solve the first of these problems. The half cycle ambiguity can be resolved by accounting for the 50 Hz data bit modulation of the C/A code. Both of these solutions, which are currently being implemented for the COSMIC program and tested on the CHAMP and SAC-C flight receivers, will reduce the N-bias to less than 1%. This remaining bias is caused by atmospheric ducting [Sokolovskiy, 2001a, 2003; Ao et al., 2003]; methods for reducing it are being investigated. Refractivity biases above 30 km can be due to either the analysis or the retrievals.

4.3. Statistical Comparisons of Temperature to Weather Analysis and Sensitivity to Boundary Condition

[32] Retrieved temperature profiles depend on the height (H_{init}) and temperature (T_{init}) used to initialize the hydrostatic equilibrium integral. In order to understand the sensitivity to these two variables, we use the CHAMP refractivities of Figure 6 and obtain seven different sets of temperature retrievals corresponding to

- Set₄₅ : $H_{\text{init}} = 45\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@45\text{km})$
- Set₄₀ : $H_{\text{init}} = 40\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@40\text{km})$
- Set₃₅ : $H_{\text{init}} = 35\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@35\text{km})$
- Set₃₂ : $H_{\text{init}} = 32\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@32\text{km})$
- Set₃₀ : $H_{\text{init}} = 30\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@30\text{km})$
- Set_{40,T+2} : $H_{\text{init}} = 40\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@40\text{km}) + 2\text{K}$
- Set_{40,T+5} : $H_{\text{init}} = 40\text{km}, T_{\text{init}} = T_{\text{ECMWF}}(@40\text{km}) + 5\text{K}$.

[33] The first 5 sets are used to examine the sensitivity of retrieved temperatures on H_{init} ; the last two correspond to artificially creating biases in the ECMWF T_{init} of +2 and +5K, respectively, in order to understand the effect of such biases on the retrieved temperatures. Figure 7 shows the means and standard deviations of CHAMP and ECMWF temperature differences for the first 5 sets. Figure 8 shows the means of temperature differences between different sets (i.e., means of Set₄₀ – Set₃₅; Set₄₀ – Set₃₂; Set_{40,T+2} – Set₄₀ and Set_{40,T+5} – Set₄₀). We make the following observations:

[34] 1. Figure 7, excluding Set₄₅, demonstrates that CHAMP and ECMWF agree to $\sim 0.5\text{K}$ in the mean and $\sim 1\text{--}2\text{K}$ (1-sigma) for individual profiles below 25 km. These results are consistent with results obtained with GPS/MET during AS-off prime periods [e.g., Kursinski et al., 1996; Rocken et al., 1997], SAC-C [Hajj et al., 2002a] and with other CHAMP analysis [Wickert et al., 2001].

[35] 2. Figure 7 indicates that the temperature means for Set₄₀, Set₃₅ and Set₃₂ are very similar. The differences between these means (Figure 8a) are $< 0.05\text{K}$ below 20 km and $< 0.02\text{K}$ for heights below 15 km and indicate that the retrieved temperatures, at least in the mean, are not very sensitive to H_{init} when chosen between 32 and 40 km.

[36] 3. Figure 8b shows that a bias in the initialization temperature will cause a bias in the retrieved mean temperature that decays exponentially with descending height at a rate of $1/e$ per $\sim 6\text{km}$ (corresponding to the atmospheric scale height). This effect can be best understood by recognizing that a bias in T_{init} at H_{init} is equivalent to a constant pressure bias at all heights below H_{init} . It follows that this effect holds true not only for the mean but also for individual occultations. From Figure 8b we conclude that a 2 K (or 5 K) bias at 40 km maps into a 0.03 K (or 0.08 K) bias at 15 km. We need to keep this effect in mind when considering potential systematic biases to RO temperatures,

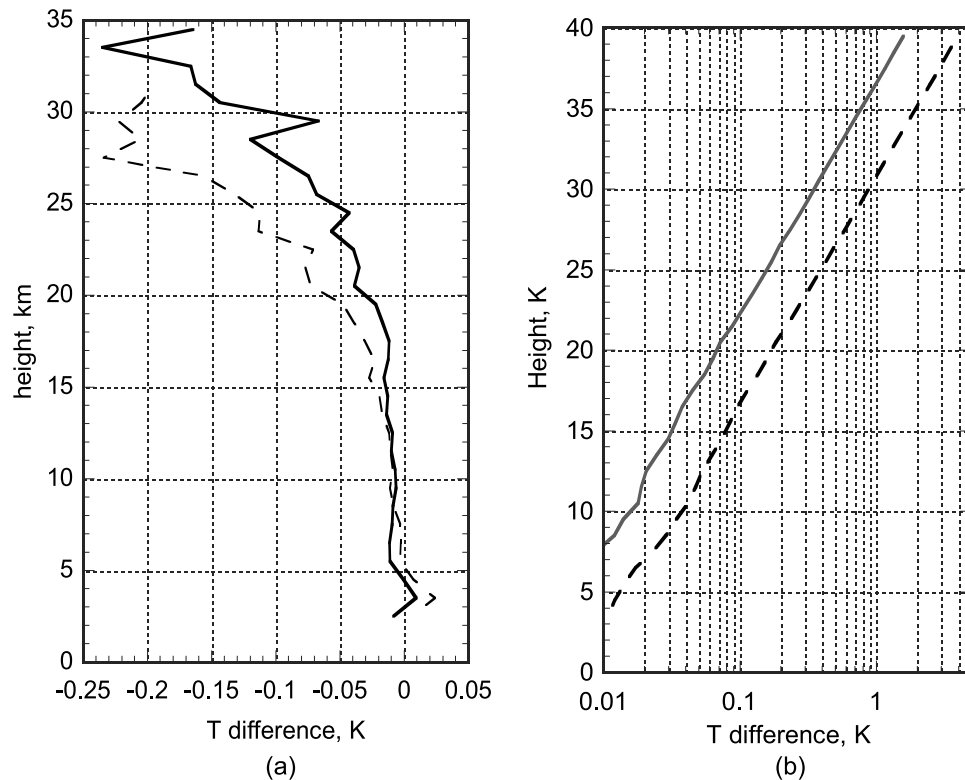


Figure 8. Sensitivity of retrieved mean temperature to height (H_{init}) and temperature (T_{init}) used to initialize the hydrostatic integral. (a) Mean temperature difference corresponding to Set₄₀ - Set₃₅ (solid line) and Set₄₀ - Set₃₂ (dashed line). (b) Same as Figure 8a but for Set_{40,T+2} - Set₄₀ (solid line) and Set_{40,T+5} - Set₄₀ (dashed line).

especially when climate-monitoring applications are concerned. This is discussed more in section 7.

[37] 4. Because of the exponential decay of temperature initialization error with descending height discussed in (3) above, ideally, one would like to choose H_{init} as high as possible (e.g., Kursinski *et al.* [1997] suggested 50 km with a nominal $T_{init} = 260$). However, large refractivity errors at higher altitudes can introduce very large temperature errors, which will propagate to lower heights. This effect can be seen clearly from the mean and standard deviation of the Set₄₅ retrievals shown in Figure 7. In that set, although the mean temperature difference is 0 at $H_{init} = 45$ km by construction, it grows to about -10 K at ~40 km because of refractivity errors at or above this altitude, causing significant temperature biases at heights well below 30 km.

[38] Observations 3 and 4 above imply that it is desirable to start as high in altitude as possible to minimize the effect of any error in T_{init} , but not too high where large refractivity noise can result in very large temperature errors. These two competing effects and the analysis presented here lead us to adopt 40 km as our initialization height whenever the ECMWF analysis is used. Another approach for optimally combining the derived refractivity data and model temperature is the 1DVAR, where no explicit H_{init} or T_{init} are needed. Instead, an effective H_{init} , which can expand over several kilometers, is introduced on the basis of the relative weighting between the assigned retrieved refractivity and model temperature errors. Further discussion of the sensi-

tivity of the temperature retrievals on the upper boundary conditions is given by Marquardt *et al.* [2003].

5. Number of Daily Occultations and Coverage

[39] A dense coverage of daily occultations and their processing in a timely manner are necessary if this type of data is to become useful for scientific and operational use. The presence of CHAMP, which carries an aft-viewing antenna, and SAC-C, which carries both fore and aft antennas, would, in principle, provide nearly 750+ occultations (250 from each antenna). In addition, the Gravity Recovery and Climate Experiment (GRACE) mission, also with fore and aft antennas, will add 500 more daily soundings, bringing the total to ~1250 profiles/d. This surpasses the number of semidaily radiosonde launches, which form a crucial input for weather analysis, and provides new data in remote areas lacking radiosonde coverage. Several operational missions are planned during the next 2–8 years, including the Constellation Observing System for Meteorology, Ionosphere and Climate (COSMIC, 6 satellites), Atmospheric Climate Experiment (ACE+, 4 satellites), National Polar-orbiting Operational Environmental Satellite System (NPOESS, 3 satellites), and the Operational Meteorology (MetOp) satellite, which is a part of the European polar orbit satellite system, in addition to several missions of opportunity, bringing the potential daily total of GPS soundings to several thousands.

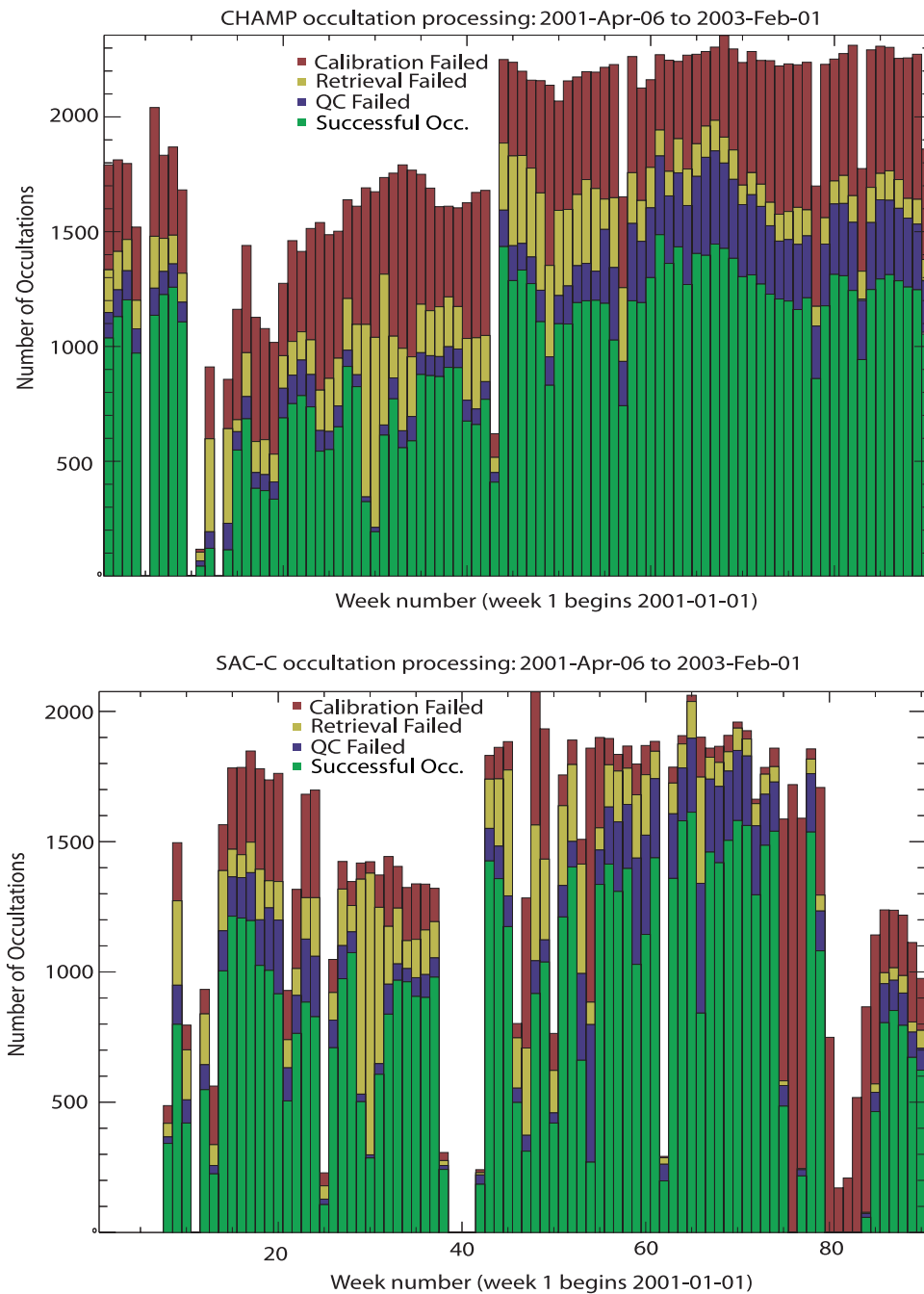


Figure 9. Total weekly number of occultation since the beginning of the CHAMP and SAC-C missions. The different colors indicate different failure modes in the processing of the occultations. The top of the red bars indicates the total number of occultation scheduled by the receiver. The top of the green bars indicates the number of those that passed all stages of the processing system including quality control. Gaps correspond to days where the receiver was not turn on.

[40] Because the forward viewing antenna on SAC-C is not yet enabled, the current target number of daily occultations is 500. Figure 9 shows the number of weekly occultations collected and processed at JPL for CHAMP and SAC-C. For an occultation to be considered successful it must pass three key processing steps: (1) calibration, where the atmospheric delay induced on the occultation link is isolated, (2) retrieval, where the delay is converted into atmospheric parameters, and (3) quality control, where key

solution parameters are checked to be within reasonable physical bounds. Figure 9 shows the number of weekly occultations failing calibration (red), retrieval (yellow), and quality control (blue), and the number passing all three steps (green). We note that the yield (number of successful occultations) has fluctuated considerably specially for SAC-C. This is due to testing various receiver software uploads and to various satellite system failures. The maximum combined number of successful occultations from

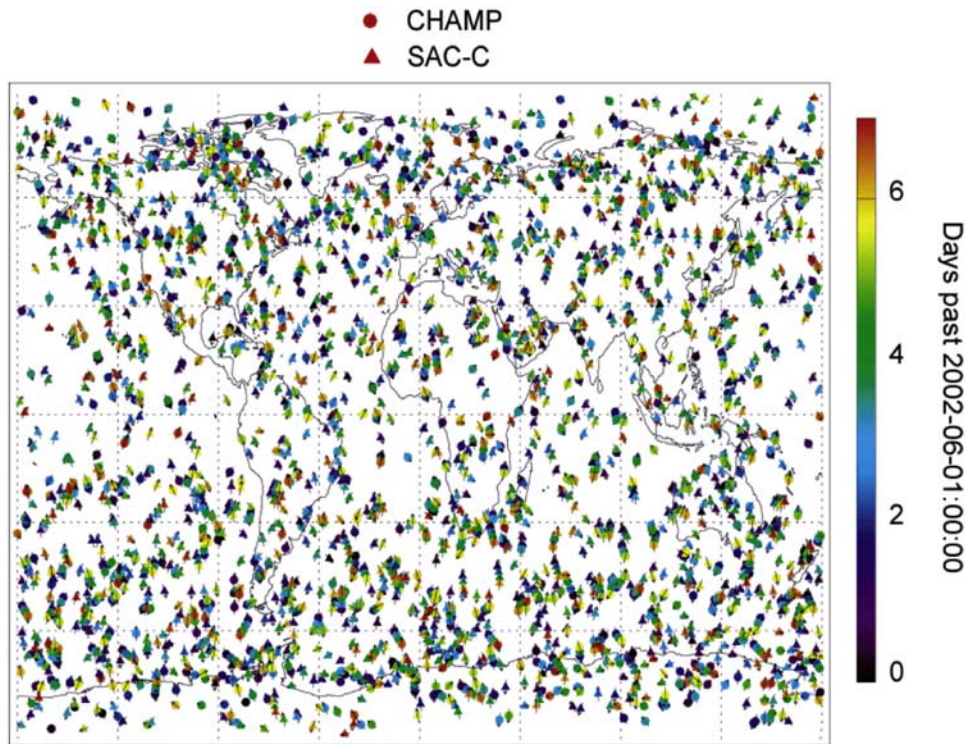


Figure 10. Map of CHAMP (circles) and SAC-C (triangles) occultation coverage for one week starting at midnight on 1 June 2002. Only occultations that passed the quality control (a total of ~ 2500 for both satellites) are shown. The color indicates the time according to the color bar spanning the period from 0 to 7 days. The arrow indicates the occultation orientation.

both satellites reaches nearly 3000/week or ~ 425 daily occultations.

[41] An example of the coverage from CHAMP and SAC-C for the week of 1–7 June 2002 is shown in Figure 10. The global nature of the measurements is evident with a lower density in the tropics than at high latitude. This is due to the high inclination of CHAMP and SAC-C and their viewing geometry relative to GPS. A LEO at low-inclination (< 30 degrees) would significantly improve the coverage near the equator. The first low-inclination satellite with GPS occultation capability will be Communications/Navigation Outage Forecasting System (C/NOFS), which is scheduled for launch in 2005 at 12 degrees inclination mainly for ionospheric sensing.

6. Sensing the Lower Troposphere

[42] Because of signal defocusing and high signal dynamics introduced by sharp structures in the lower troposphere, tracking the occulted GPS signal in that region has proven to be difficult. For example, our experience with GPS/MET has shown that very often the GPS signal is lost in the lower troposphere in regions of high water vapor concentration. An exception was a period during June–July 1995 where the signal penetrated to the lowest 2 km in the majority of cases [Kursinski and Hajj, 2001]. That improvement resulted from an experimental mode of operation in the receiver known as “flywheeling.” That same mode, with some enhancements, is on CHAMP and SAC-C, allowing profiles to penetrate the lowest 0.5 km of the atmosphere in the majority of cases.

[43] The left panel of Figure 11 shows the percentage of occultations that reach below a certain height for CHAMP (a similar value is obtained for SAC-C) at different latitude bands. The figure also shows the penetration of occultations from the GPS/MET flywheeling period (middle panel) and a GPS/MET period with no flywheeling (right panel). We see that penetration into the lowest 1/2 km over the entire globe has gone from $\sim 5\%$ (no flywheeling) to 30% (GPS/MET flywheeling) to $\sim 60\%$ for CHAMP. Further improvement is expected with the implementation of “open-loop” tracking on CHAMP and SAC-C.

[44] A number of issues complicate tracking in the troposphere. We can illustrate these with an occultation acquired by CHAMP on 1 October 2001. We start by examining the SNR for this occultation (Figure 12), which shows a disappearance of the signal at 40 s into the occultation and then its reappearance nearly 8 s later for a few seconds before it finally disappears. This temporary “blocking” of the signal is caused by atmospheric ducting or strong super refraction when the signal is trying to penetrate the region immediately below the height of sharp refractivity gradient [Hajj *et al.*, 1994; Sokolovskiy, 2003]. The corresponding partial water vapor pressure (WVP) profiles obtained from this occultation are shown in Figure 13, along with NCEP analysis and other radiosondes measurements. Because of the relatively small sea surface temperature variation, structures over oceans can be correlated for distances extending over 1000 km. The nearest radiosonde to the occultation found, Esperance, is nearly 600 km away from the occultation and shows resemblance to the radio optics (RO) retrieval.

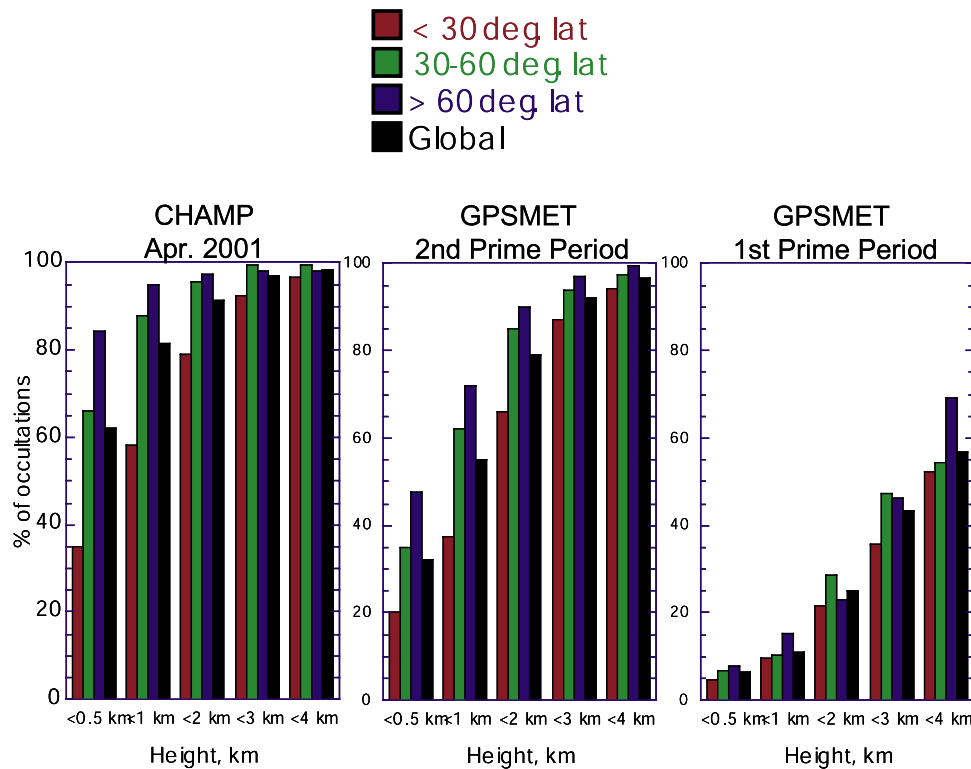


Figure 11. Percentages of occultations penetrating below a specified height for different latitude bands and globally. This is shown for (left) CHAMP on the basis of 900 early CHAMP occultations, and compared to the (middle) second and (right) first prime periods from GPS/MET.

[45] The CHAMP WVP of Figure 13 are obtained by assuming the NCEP temperature. Two solutions for the same occultation are shown corresponding to two different retrieval schemes: (1) the canonical transform (CT) solution [Gorbunov, 2002], which maintains the highest possible vertical resolution and overcomes some difficulties associated with atmospheric multipath, and (2) the radio optics (RO) solution, which relies on fitting the spectrum of the occulted signal [Gorbunov, 2001]. While there are similarities in the two solutions (e.g., sharp change around 1.9 km and a local maximum above this height), there are significant differences that point to the difficulties associated with sensing the lower troposphere. A major difficulty is the negative N-bias evident in the smaller WVP of the CT solution when compared to the RO solution and the NCEP analysis. The uncertainties associated with retrievals in the lower troposphere and finding a most accurate inversion scheme in that region is an open area of research.

6.1. Flywheeling

[46] Without flywheeling, the receiver would lose the signal once the SNR drops below about 35, which, in the above example, would end the occultation after ~ 40 s (Figure 12) and lose all information about the structure below 2 km. In flywheeling, when $\text{SNR} < \text{SNR}_{\text{cutoff}}$ ($= 35-50$), the receiver extrapolates the model for the signal Doppler shift and Doppler shift rate (to maintain its third-order phase-lock loop) on the basis of data from the previous few seconds. The receiver will then record the real and imaginary correlation sums, on the basis of which the SNR and phase difference between the received signal and the extrapolated model are

computed [Thomas, 1995]. (These sums are the real and imaginary sums of the product of the received signal sampled at 20 MHz and an internally generated model of the coarse acquisition (C/A) code of the GPS satellites. The sums are over 0.02 or 0.01 s, corresponding to the sampling interval.) This continues until the $\text{SNR} > \text{SNR}_{\text{cutoff}}$ (after which the receiver goes back into close-loop mode) or 15 s have elapsed (after which the occultation is ended). During flywheeling, if the difference between the modeled (based on extrapolation) and true average atmospheric Doppler shift, Δf , stay well within the receiver's sampling rate window (± 25 Hz for this example), then it would be possible to extract the true phase and amplitude of the occulted signal. When Δf is outside the ± 25 Hz window, two things happen: Aliasing makes it hard to recover the true phase and amplitude and the signal is significantly weakened by the $\sin(\pi\Delta f T_s)/\pi\Delta f T_s$ filter introduced by the receiver, where T_s is the sampling time ($= 1/50$ s).

6.2. Open Loop Atmospheric Doppler Model

[47] To further understand the dynamics of the signal and examine the degree of success of tracking it in the lower troposphere, occultation data were sampled at 100 Hz from SAC-C during 5 November 2001 to 25 January 2002. (The normal rate is 50 Hz for CHAMP and SAC-C.) When sampled at 100 Hz, the signal is recoverable if the flywheeling model remains within ± 50 Hz from the true average atmospheric Doppler shift. Using a procedure suggested by Hajj et al. [2002b], it is possible to predict the atmospheric bending, α , (and therefore the Doppler shift) based on knowledge of Θ , which is the separation

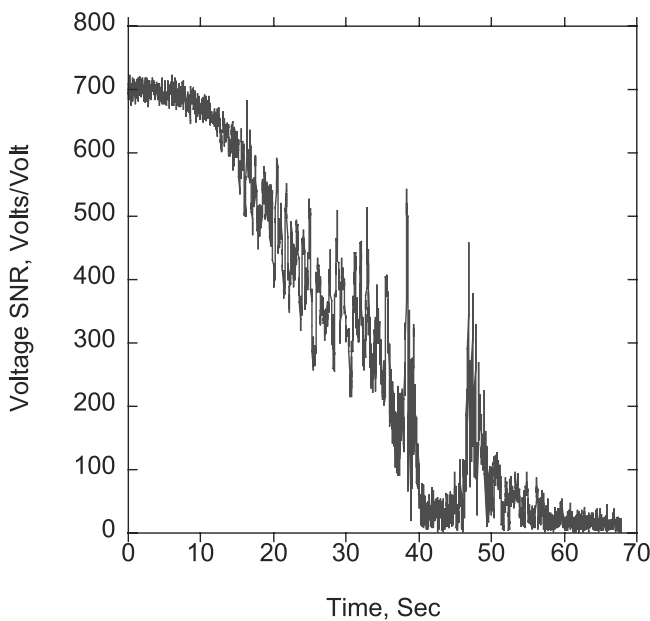


Figure 12. Signal-to-noise ratio of a CHAMP occultation (with ID = 2001-10-01-02:19champ_gps23) as a function of time showing the disappearance of the signal for a duration of ~ 8 s commencing at the time when the occulted signal is immediately below the top of the planetary boundary layer. The signal reemerges when the condition for total internal reflection ($dN/dz < -157$) is no longer satisfied.

angle between the occulted GPS and LEO satellites relative to the center of curvature of the geoid in the vicinity of the occultation tangent point (Figure 14a).

[48] The functional form of $\alpha(\Theta)$ depends very weakly on atmospheric conditions (which is why it can be used for predicting α) but is very strongly dependent on the LEO radius and therefore has to be recomputed for different LEO satellites. The function $\alpha(\Theta)$ is computed by fitting a series of occultation data from a LEO satellite and therefore implicitly contains all the atmospheric bending effect. This procedure provides a Doppler shift prediction that is accurate to better than 10 Hz for LEOs with nearly circular orbits such as CHAMP and SAC-C. A different approach, which relies on an atmospheric model and ray tracing for deriving $\alpha(\Theta)$, is suggested by Sokolovskiy [2001b] and has the advantage of being accurate (~ 10 Hz) for arbitrary orbits.

[49] Using the predicted Doppler shift based on knowledge of $\alpha(\Theta)$, we examine how well the “flywheeling” is performing. Figure 14b shows the difference of measured (after 1-s averaging) and modeled (based on the bending of Figure 14a) Doppler shifts as a function of the straight-line tangent point for 99 SAC-C occultations taken on 6 November 2001. Deviations of order ± 10 Hz from the model reflect true structure in the atmosphere. Larger deviations can be attributed to tracking loop errors; however, as long as the deviation does not exceed $\pm$ half the receiver’s sampling rate (± 50 Hz), the phase and amplitude can be considered to be recoverable.

6.3. Atmospheric Ducting and Multipath

[50] When several tones are simultaneously observed, careful interpretation of the signal is needed to obtain

accurate retrievals. As an example, these multiple tones can be seen in the power spectra (Figure 15) of the occultation of Figure 12. These spectra are obtained by subtracting the atmospheric phase as recorded by the receiver from the modeled phase (as described above), and then taking the complex fast Fourier transform (FFT) with a 2.56 s moving window (corresponding to 128 points at 50 Hz). The following features are seen in the spectra: (1) One tone is dominant at the beginning of the occultation until 35 s. The alternating dark and bright horizontal bands between 0 and ~ 35 s correspond to narrow and broad spectra around the main tone, respectively. This narrowing and broadening of the spectra is due to fitting an integer (for narrow) and an integer and a half (for the broad) number of cycles within the 2.56 s window used to do the FFT. (2) The signal splits into 2 or 3 tones between 35 and 38 s, disappears between 38 and 46 s because of ducting and reappears with broad spectra (5–6 Hz) between 45 and 50 s. (3) A tone appears at ~ 30 s and crosses the figure diagonally. This tone can be seen with different intensity in many occultations. It was first observed in LEO to geostationary satellite occultations and interpreted as a surface reflection by Pavelyev *et al.* [1997]. Later observation from GPS/MET and simulation of the ocean reflected signal Doppler shift relative to the direct signal confirmed this interpretation [Beyerle and Hocke, 2001]. A more in depth analysis of this ocean/ice reflected signal detected in the CHAMP occultation data was given by Beyerle *et al.* [2002]. (4) A horizontal spread of power is observed at 25 s, which is due to the receiver slipping several cycles at that time.

[51] The various tones present in the spectra are due to atmospheric multipath caused by the sharp refractivity gradient such as encountered around the tropopause and in the lower troposphere. This and similar spectra show that a faint signal continues to exist even after the direct and the reflected signals merge. There are two reasons for this: (1) In the presence of a sharp refractivity gradient, the last signal

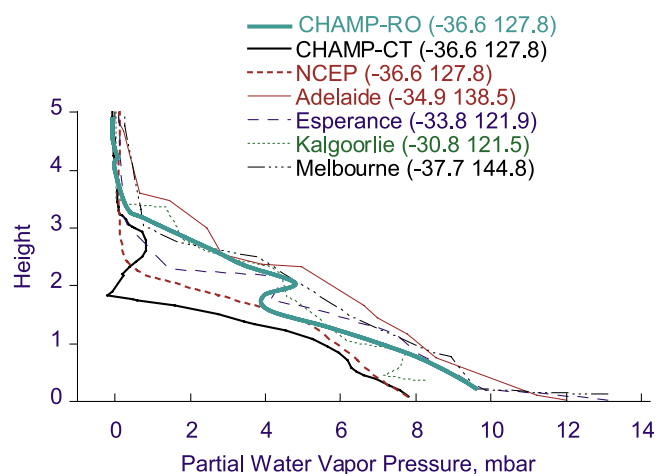


Figure 13. Water vapor partial pressure of the occultation of Figure 12 and corresponding NCEP and nearby radio-sonde profiles. The NCEP temperature is used to derive partial water vapor pressure from CHAMP refractivity. Two different CHAMP retrievals are shown corresponding to the canonical transform (CHAMP-CT) and radio optics (CHAMP-RO) solutions.

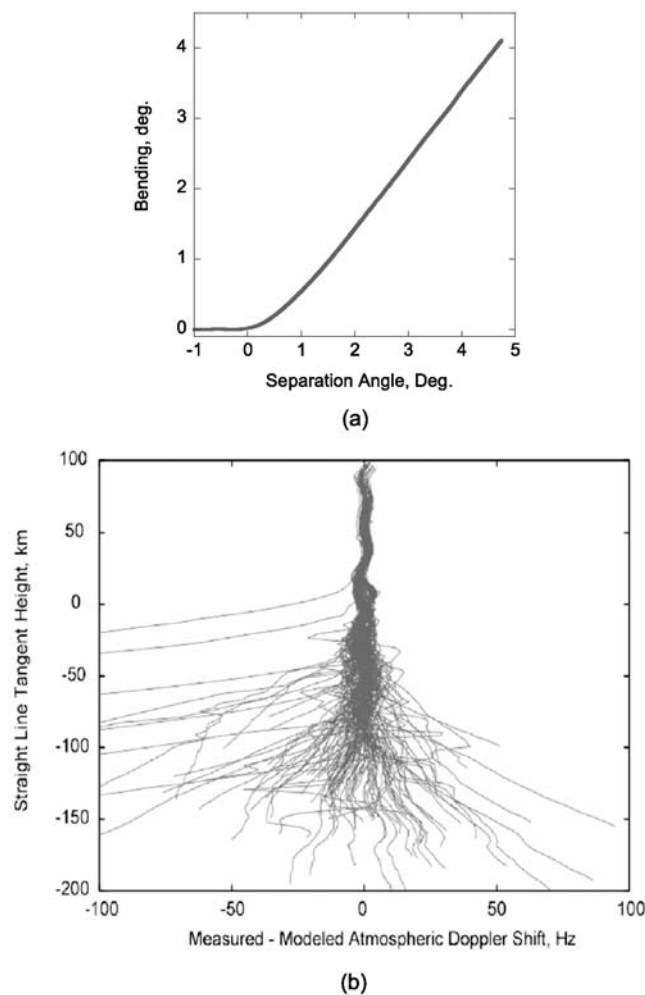


Figure 14. (a) Atmospheric bending as function of separation angle between the transmitter and the receiver showing the strong linear relationship between them in the troposphere, which can be exploited to derive a model for the Doppler shift of the occulted signal. (b) Difference between the measured and modeled Doppler shift for all SAC-C occultations collected in 1 day at the 100 Hz rate. The Doppler shift model is based on Figure 14a. Deviations larger than ± 50 Hz are indicative of episodes where the receiver is not tracking properly.

detectable by the receiver is not coming from near the Earth's surface but rather reflected internally from the height where the refractivity gradient is sharp. (2) Often a faint tone continues to exist even when all signals are completely gone because of not accounting for the 50 Hz data bit modulation of the C/A code, therefore, limiting the derived phase to be in the $[-\pi/2, \pi/2]$ rather than the $[-\pi, \pi]$ range. This causes even white noise to have a finite power spectrum. This problem will disappear once the data bit modulation is accounted for.

6.4. Sensing the Boundary Layer

[52] Figure 16 presents another example of sensing the lower troposphere by a CHAMP occultation collected near 45°N and 70°W (over Maine) on 17 November 2001. The recorded SNR for this occultation (Figure 16a) shows a

temporary fading for about 10 s during the occultation, although not as strong as that of Figure 12 indicating that super refraction condition is not met here. For this occultation a high-resolution radiosonde sounding taken within 2 hours and 100 km south of the occultation was found and used for comparison. The CHAMP refractivity profile (obtained using CT), along with those from the radiosonde and ECMWF analysis are shown for the lowest 7 km in Figure 16b. Using the CHAMP refractivity and ECMWF analysis temperature, we derive the CHAMP specific humidity, which is shown in Figure 16c, indicating a strong correlation with the sonde measurements. The sharp temperature inversion at ~ 1 km and the constant virtual potential temperature below 1 km evident in the analysis and the sonde measurements (Figures 16d and 16e) are indicative that the sharp gradient in the specific humidity (Figure 16c) is associated with the top of the planetary boundary layer (PBL) at ~ 1 km. The negative CHAMP specific humidity is due to the underestimate of temperature by the ECMWF analysis, which is confirmed in Figure 16d. The sharp refractivity gradient associated with the top of the PBL intensifies atmospheric effects and causes abrupt weakening (or disappearance in the case of super refraction)

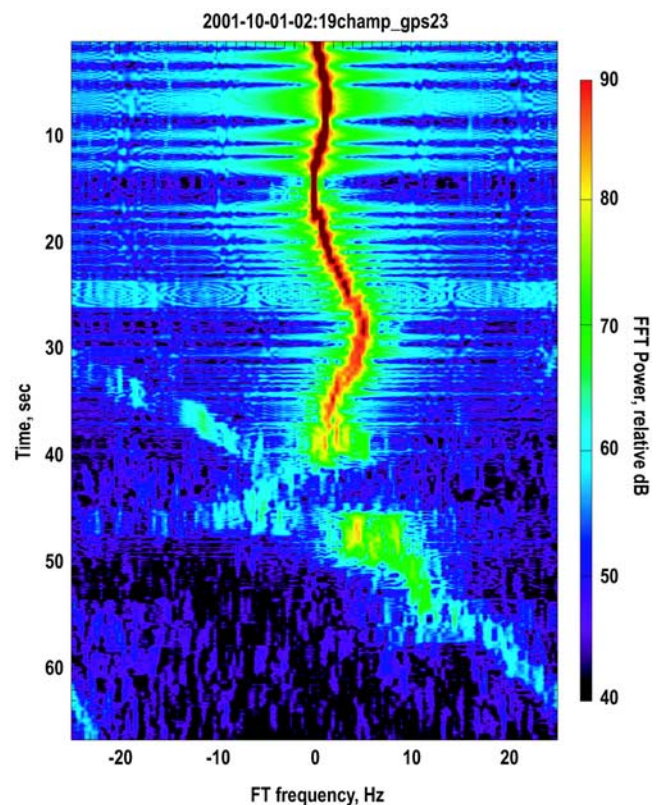


Figure 15. FFT of the complex (real and imaginary) difference between the received and modeled signal of the occultation in Figure 12. The FFT is computed on the basis of a 2.56 s moving window. Immediately before and after the signal gap (between 38 and 48 s), multiple tones can be seen as evidence of atmospheric multipath. The signal running diagonally starting at ~ 30 s corresponds to the occulted signal reflected off the ocean's surface.

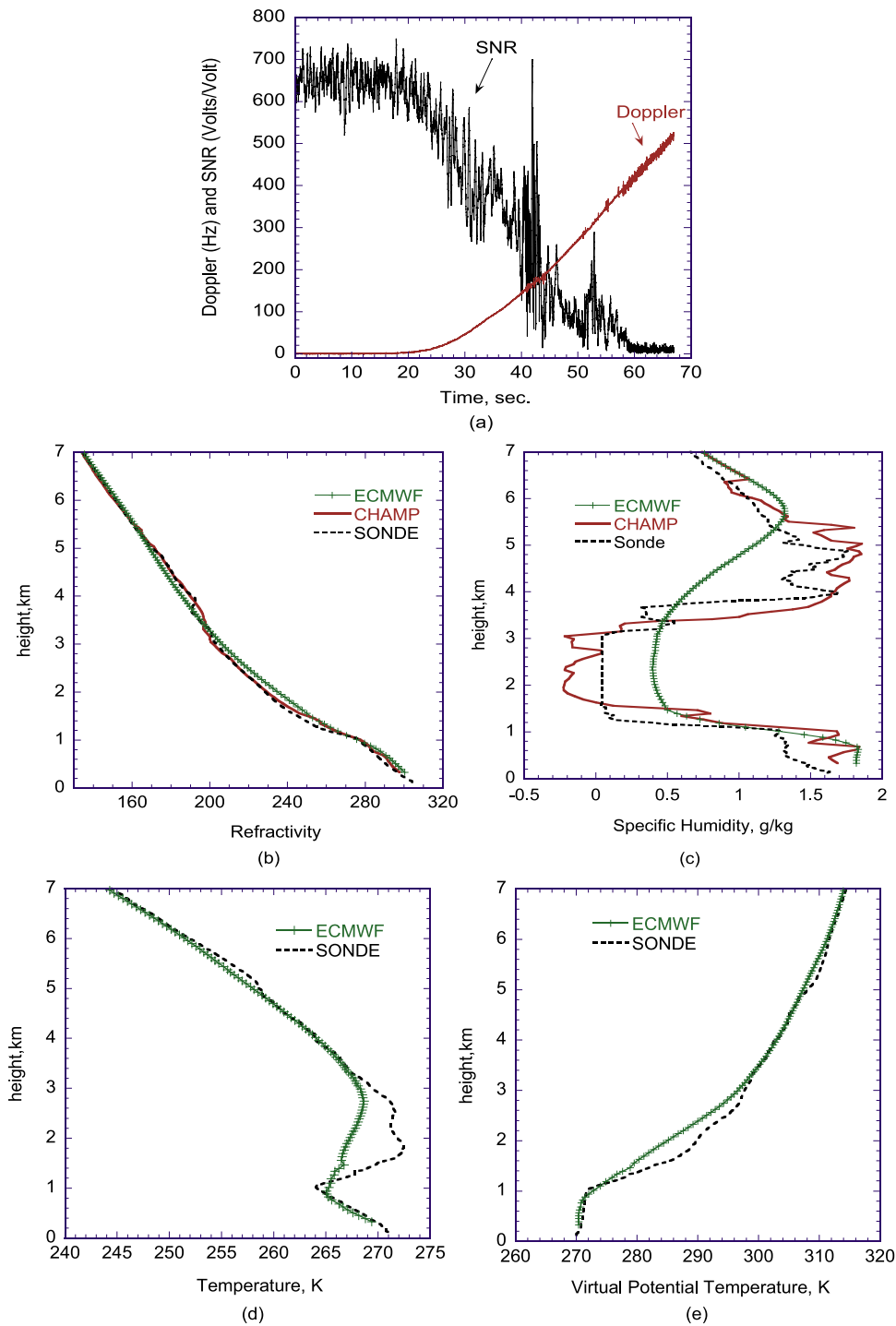


Figure 16. SNR and atmospheric Doppler for a CHAMP occultation (with ID = 2001-11-17-11:05champ_gps13) at 45°N and 70°W (Figure 16a) and corresponding refractivity (Figure 16b) and specific humidity (Figure 16c) derived by assuming the ECMWF temperature. Refractivity, specific humidity, temperature and virtual potential temperature from a nearby radiosonde and ECMWF analysis are shown in Figures 16b–16e, respectively.

of the signal intensity, therefore, potentially providing information on the height of the PBL.

7. CHAMP–SAC-C Cross-Comparison

[53] The presence of two missions concurrently collecting occultations provides the first opportunity to examine the

precision of these measurements and potentially assess the technique's claim to sub-K temperature accuracy. Even though, in the strictest sense, comparison of CHAMP and SAC-C will establish only a level of consistency between the two measurements, their often very different viewing geometries can cause them to be subject to quite different atmospheric and ionospheric propagation, asymmetry, and

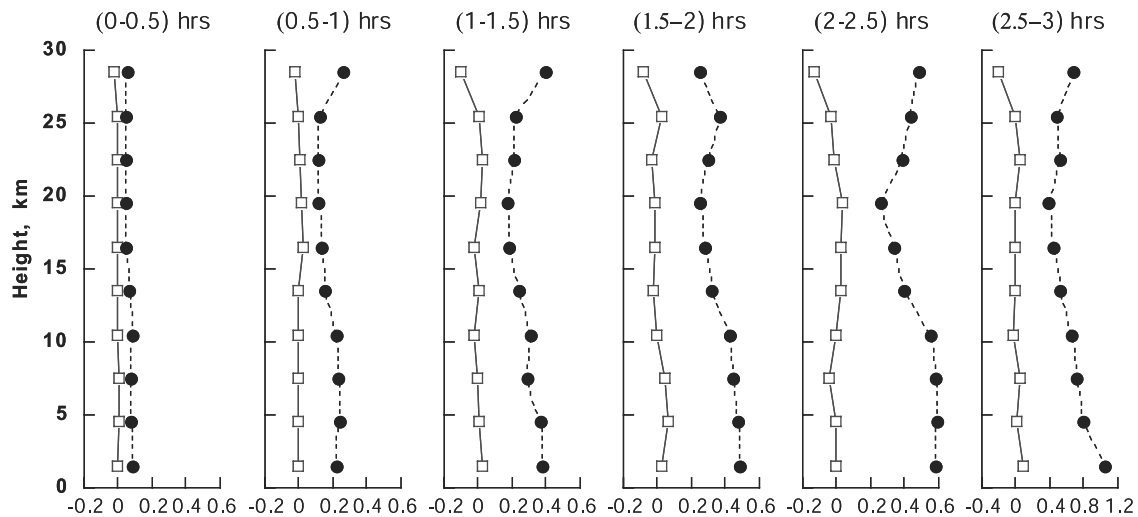


Figure 17. Mean (squares) and standard deviation (circles) of temperature differences (in degrees) as a function of height between two sets of profiles: (1) NCEP temperature at CHAMP occultation location and time and (2) NCEP temperature at CHAMP occultation location and SAC-C occultation time. The statistics are binned according to the time separation between CHAMP and SAC-C to provide a measure of temporal variation of the atmosphere according to the NCEP analysis.

mismodeling effects, which can be a major source of error. Careful study of coincident soundings, particularly the closest in space and time, can reveal a good deal about overall accuracy. Many sharp or unusual features observed in common by the two, from different viewpoints, can only be interpreted as real features in the atmosphere and are further indicative of accuracy.

[54] In searching for coincident CHAMP and SAC-C occultations, we first consider all the data collected from these two satellites in the period 22 August 2001 to 15 October 2001 with a total of ~ 4200 CHAMP and ~ 6800 SAC-C occultations. In collocating CHAMP and SAC-C occultations, we must first decide on the proper time and space windows appropriate for two occultations to be considered “close.” In determining these windows we first try to answer the following two questions: (a) How much does the atmosphere vary with time at a fixed location? (b) How much does it vary in space for a fixed time?

7.1. Atmospheric Variation in Time

[55] We answer question a by the following procedure: (1) Find all CHAMP and SAC-C occultation pairs that are within 800 km distance and 3 hours from each other. (For the moment the distance is approximately determined on the basis of the position of the tangent point at 30 km altitude.) The number of these pairs is ~ 1400 . (2) For each pair, find the NCEP profile at the location of the CHAMP occultation at the two different times corresponding to the CHAMP and SAC-C occultations (a linear interpolation between the 12 hourly NCEP analyses is used). (Because the duration of an occultation is ~ 60 s, it is fair to assume that the atmosphere is not changing in any significant way during this time and to assign a single time to each occultation.) (3) We divide the occultation pairs in 1/2 hourly time bins and we form statistics for each bin. The mean and standard deviation of the temperature difference of all pairs within each bin are computed as a function of height and shown in Figure 17. This figure shows that changes of order 0.5 degrees are expected within

1.5 hours. Because of the low time resolution of the model (12 hours), we expect this estimate to be a lower bound of the real variability in the atmosphere; therefore, in answering question b above, we limit ourselves to occultations pairs that are in the first bin (i.e., $<1/2$ hour apart).

7.2. Atmospheric Change With Distance

[56] In order to answer question b above, we consider the pairs of CHAMP-SAC-C occultations that are $<1/2$ hour apart (a total of ~ 300). Each pair is represented graphically by a line in Figure 18 showing the distance between the two occultations as a function of altitude (computed on the basis of the estimated location of the tangent point for each occultation at that altitude). Next, for each pair, we obtain the corresponding NCEP profiles at the times and locations of the CHAMP and SAC-C occultations and compute the temperature difference between them as a function of altitude. Figure 19 is a scatterplot of the NCEP temperature differences obtained this way as a function of distance for altitudes below 30 km. Because of the very small change in temperature introduced by the model within 1/2 hour (left-most panel of Figure 17), temperature changes in Figure 19 are mainly due to variation with distance. The root-mean-square (RMS) temperature difference in Figure 19 is less than 0.65 K for distances less than 200 km. As in the time variation estimate above, we expect this to represent a lower bound on the true gradient that can be observed in the atmosphere.

[57] In summary, on the basis of the NCEP analysis, temperature changes in the atmosphere within 1/2 hour and 200 km radius are ~ 0.66 K or more. Below we examine pairs of profiles within this time-space window in more detail.

7.3. Comparison of Profiles Within 1/2 Hour and 200 km Window

[58] All retrievals used in subsequent analysis are obtained by initializing the hydrostatic integral at 40 km with temperature obtained from the ECMWF analysis. For the period 22 August 2001 to 15 October 2001 there

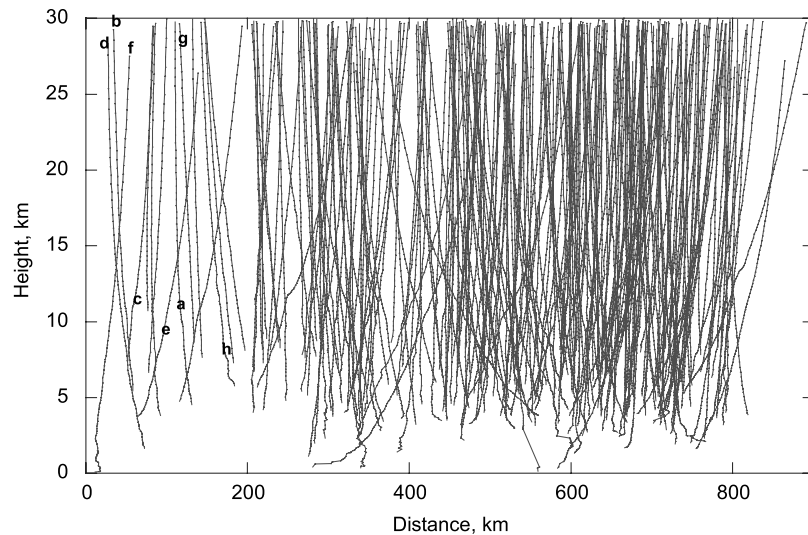


Figure 18. Each line corresponds to a CHAMP-SAC-C pair of occultations that are <0.5 hours and <800 km apart and shows the distance between tangent points as a function of height. All occultations from 22 August 2001 to 15 October 2001 satisfying the condition above are shown. Marked a–h are the nearest pairs shown in Figure 20. The lower limit of each occultation corresponds to a height where the retrieved $T = 250$ K.

are only 15 profiles that are within 1/2 hour and 200 km apart as seen in Figure 18. Of these 15, we examine 8 in detail selected at different latitudes with noteworthy features. Figures 20a–20h correspond to the lines marked a–h in Figure 18. Each example in Figure 20 also shows the ECMWF TOGA analysis interpolated to the locations and times of CHAMP and SAC-C, the average latitude and longitude of the occultation pair, the average orientation, and the IDs of the occultations involved. Starting with the high-northern-latitude region, Figure 20a shows close agreement between the CHAMP and SAC-C profiles at

nearly all heights, but especially near the tropopause. On the other hand, the temperature structure around the tropopause is conspicuously different for the ECMWF analysis, which tends to smooth sharp structures and thus overestimate the temperature at the tropopause. In this example, because the differences between CHAMP and SAC-C are <1 K everywhere below 20 km altitude, while it is significantly larger (~ 2 K) between the occultation profiles and the analyses around and above the tropopause, the discrepancy is very likely due to smoothing biases in the analyses. The close agreement

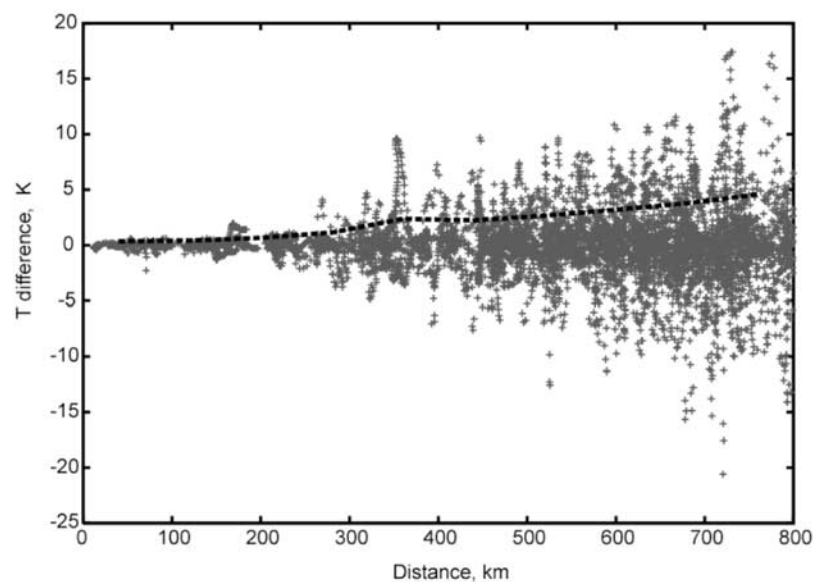


Figure 19. Temperature difference as a function of distance between pairs of NCEP profiles obtained at the coincident occultations of Figure 18. The RMS difference is indicated by the solid dashed line.

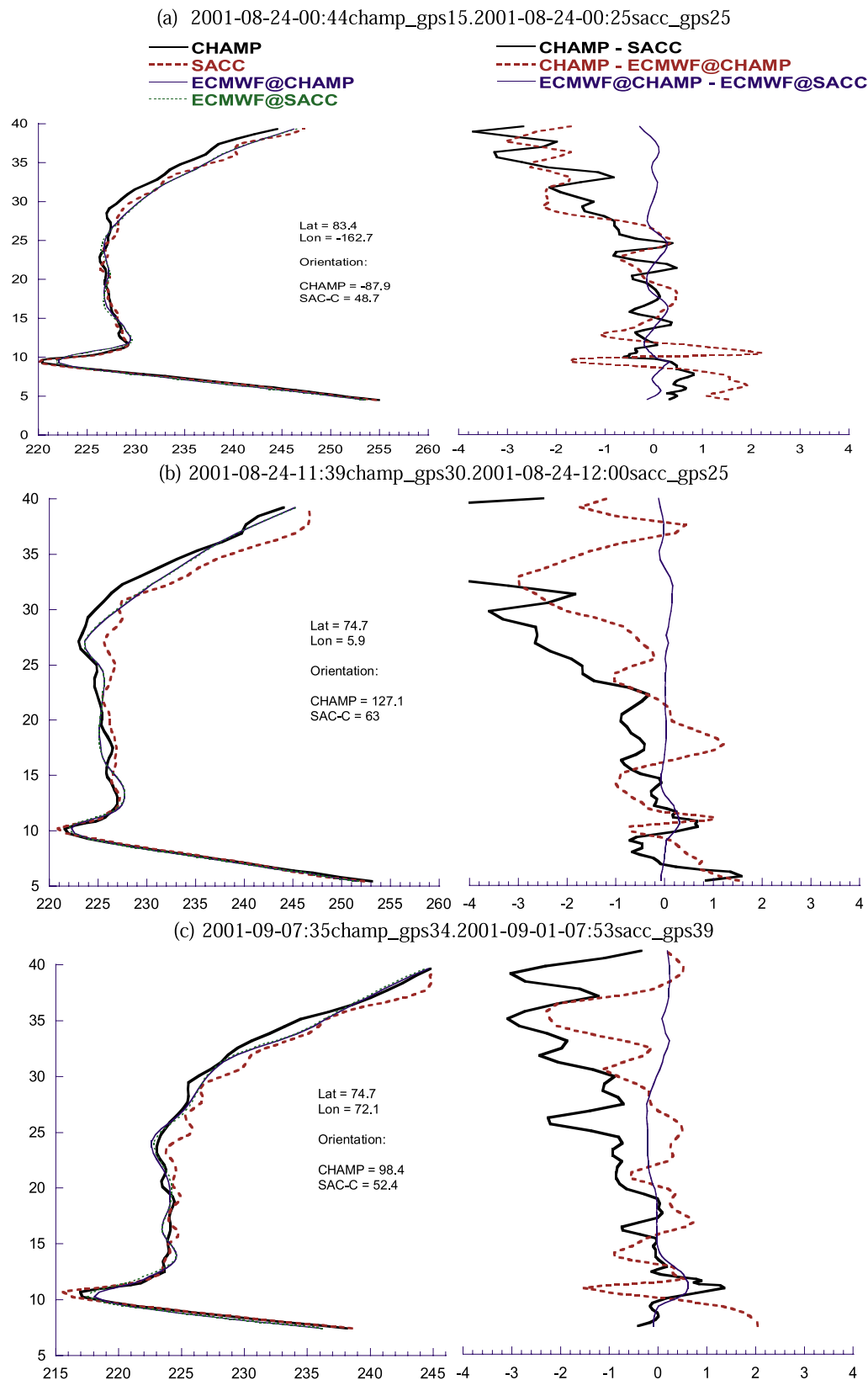


Figure 20. Examples of temperatures (left panels) and temperature differences (right panels) of CHAMP and SAC-C occultations that are <1/2 hour and <200 km apart for the period 22 August 2001 to 15 October 2001. Figures 20a–20h correspond to the a–h labels in Figure 18. Also shown is the ECMWF analysis at the CHAMP and SAC-C locations and times, the average latitudes and longitudes of the occultations, the orientations of CHAMP and SAC-C occultation links (measured in degrees counterclockwise from East), and the IDs of the pair of occultations compared.

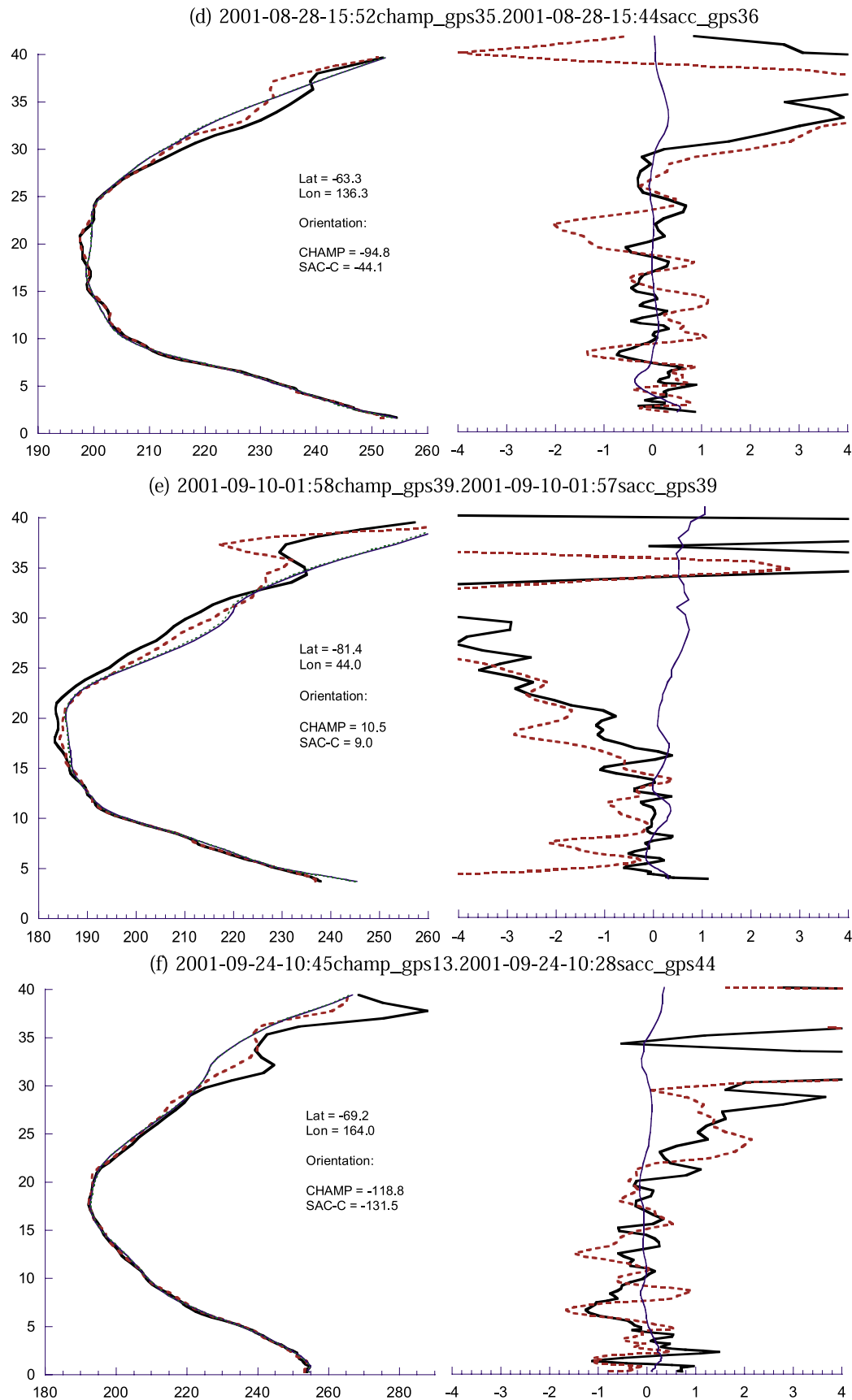


Figure 20. (continued)

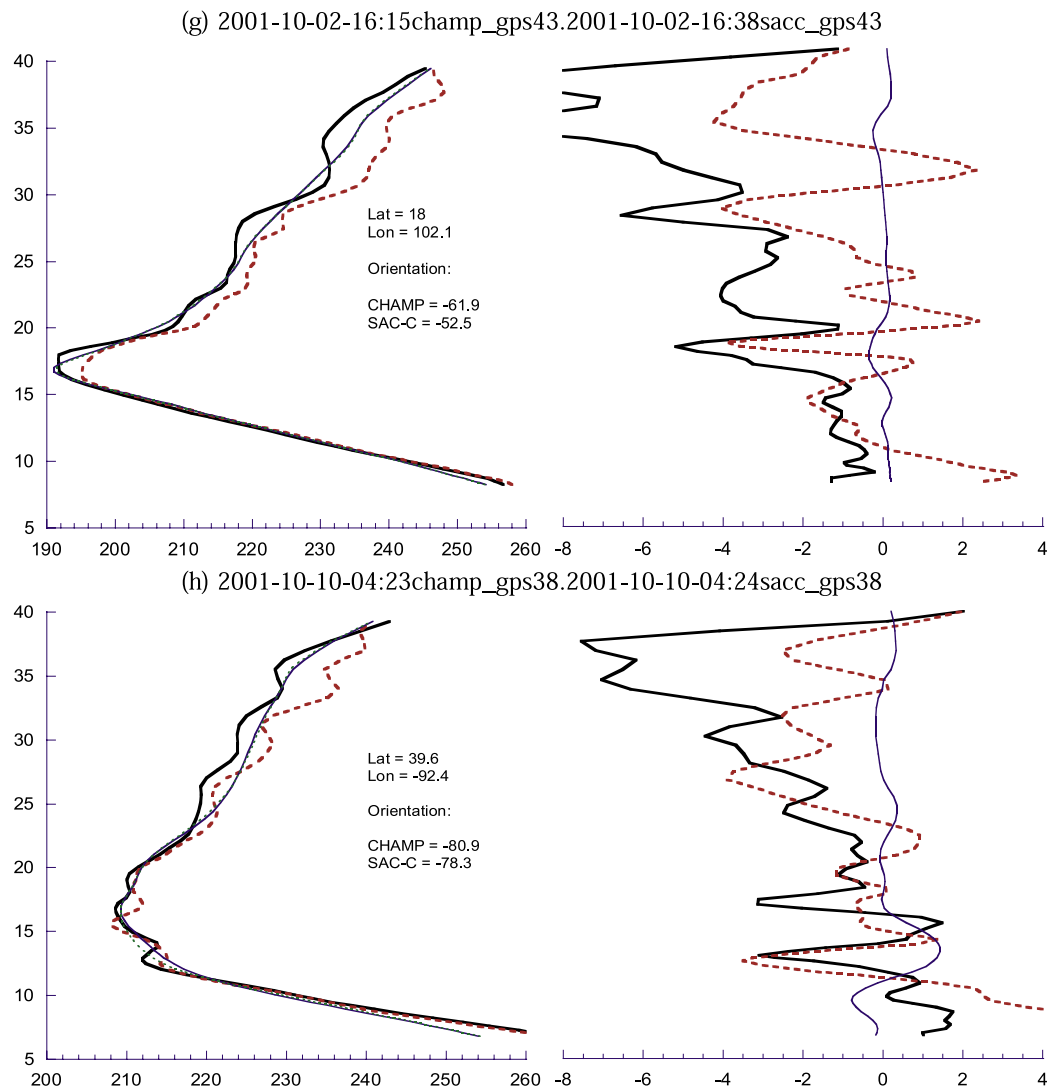


Figure 20. (continued)

between CHAMP and SAC-C, different instruments with different antennas and viewing geometries, tends to support expectations based on early theoretical studies [Kursinski *et al.*, 1997].

[59] Similar features are seen in the examples of Figures 20b and 20c taken at high northern latitudes. As in the example of Figure 20a, the analyses are smoothing the structure around the tropopause resulting in an overestimate (underestimate) below (above) the tropopause. We note that in Figure 20b the difference between the CHAMP and SAC-C temperatures at the tropopause is reflected in the temperature difference between the corresponding ECMWF profiles at the same altitude, suggesting that the gradient could very well be real. A similar gradient is seen in the occultation pair and the analyses in the example of Figure 20c. We further note that there is a significant correlation between the CHAMP and SAC-C wavy structures in the stratosphere in both examples of Figures 20b and 20c suggesting that these waves are real. In all examples of Figures 20a–20c, the lapse rates above and below

the tropopause are significantly larger in the occultation profiles than in the analyses.

[60] Moving to high southern latitudes, we show the three examples in Figures 20d–20f. Some of the common features are as follows: (1) CHAMP and SAC-C are significantly closer to each other than they are to the analyses at heights between 5 and 15 km. (2) Temperature lapse rates agree very well between CHAMP and SAC-C in the troposphere but deviate significantly from the analyses (Figures 20d and 20e below 10 km). (3) Sudden changes in the temperature lapse rate reflecting the higher vertical resolution, which is missed by the analyses. These changes match almost exactly between CHAMP and SAC-C (e.g., structures at ~2 km and 5 km in Figure 20d and at ~7 km in Figure 20e). (4) The CHAMP and SAC-C temperatures near the surface in Figure 20e (the Antarctic surface is at ~4 km altitude) and Figure 20f are significantly lower than indicated by the ECMWF analyses. (In Figure 20f, the somewhat jagged CHAMP and SAC-C temperature structure is most likely due to the dry-wet ambiguity present when the

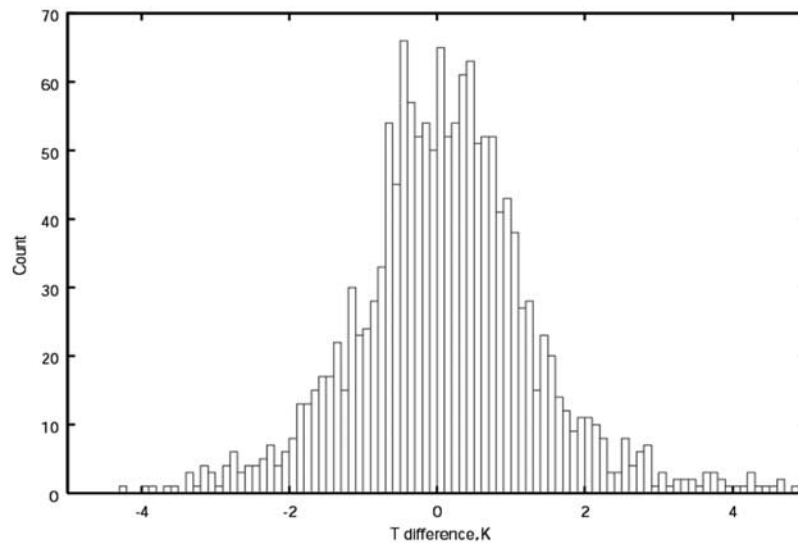


Figure 21. A histogram of CHAMP and SAC-C temperature differences for all points in occultation pairs that are between 5 and 10 km altitude and within 150 km from each other.

temperature is higher than 250 K.) (5) In the stratosphere, there is a clear correlation in the wavy structure captured by CHAMP and SAC-C and a significant deviation from the analysis.

[61] We finally consider two examples at low latitudes (Figure 20g) and midlatitudes (Figure 20h). In Figure 20g we see a significant temperature gradient between the CHAMP and SAC-C occultation near the tropopause that is absent in the analyses. If this gradient is real, it would be hard to capture with the analysis since the two occultations are only ~ 120 km apart (Figure 18), or within one grid in

the analyses. Given the level of accuracy of GPS occultation near the tropopause, as suggested by previous examples, the observed gradient is likely to indicate a real temperature gradient in the atmosphere.

7.4. Statistical Comparison

[62] To quantify the difference between CHAMP and SAC-C temperature measurements in a statistical sense, we expand our search for coincident occultations to the period covering 10 July 2001 to 29 March 2003. We limit our search to occultations that satisfy both of the following

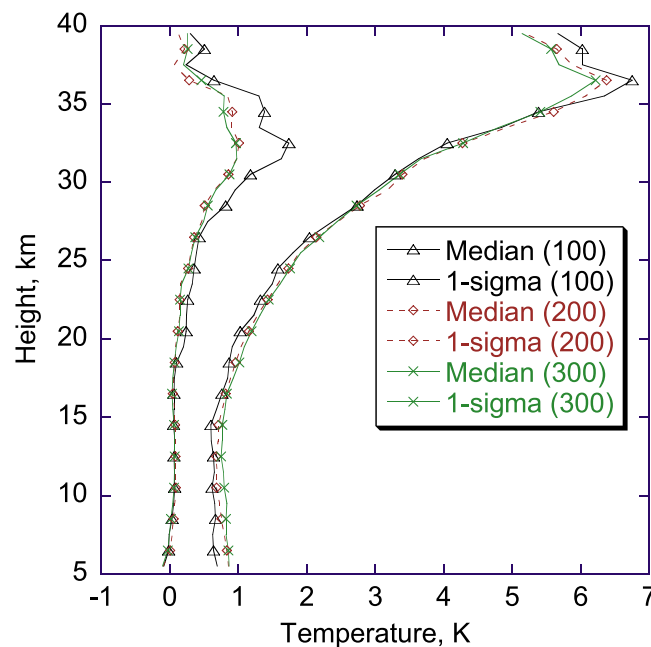


Figure 22. Measures of consistency between the CHAMP and SAC-C as a function of height for measurements $< (100, 200, 300)$ km apart. The median corresponds to the LHS of equation (6b), and the 1-sigma corresponds to the square root of the LHS of equation (6a).

criteria: (1) They occur less than 1/2 hour apart and (2) the tangent points of the two occultations are less than 200 km at least for one height. (The latter condition implies that some points are more than 200 km apart.) The data searched contain ~67,000 CHAMP and ~62,000 SAC-C occultations that were successfully retrieved. The number of pairs found that satisfy the two criteria mentioned above is 212. Combining these 212 pairs of coincident occultations into a set S , each point in S is characterized by the following:

- i index of the CHAMP-SAC-C pair (1 to 212);
- h a tangent point height;
- $\vec{x}_{1,i}(h), \vec{x}_{2,i}(h)$ positions of a tangent point at height h for CHAMP and SAC-C, respectively, for pair i ;
- $d_i(h)$ distance ($|\vec{x}_{1,i} - \vec{x}_{2,i}|$) between tangent points;
- $t_{1,i}, t_{2,i}$ times of the CHAMP and SAC-C occultations, respectively, for pair i ;
- $T_{\text{champ},i}(h)$ CHAMP retrieved temperature at height h , location $\vec{x}_{1,i}$ and time t_1 for pair i ;
- $T_{\text{sac-c},i}(h)$ SAC-C retrieved temperature at height h , location $\vec{x}_{2,i}$ and time t_2 for pair i ;
- $T_{\text{ecmwf@champ},i}(h)$ ECMWF temperature at $h, \vec{x}_{1,i}$ and $t_{1,i}$;
- $T_{\text{ecmwf@sac-c},i}(h)$ ECMWF temperature at height $h, \vec{x}_{2,i}$ and $t_{2,i}$.

[63] In order to infer the precision of temperature from CHAMP and SAC-C, we examine a histogram of ΔT_{gps} for all points in S satisfying $d < 150$ km and h between 5 and 10 km (Figure 21). To obtain robust statistics less sensitive to outliers (due, for example, to sharp temporal or spatial temperature variation), we use the median and the 68% confidence interval as measures of the mean and standard deviation in ΔT_{gps} . The 68% confidence interval is defined as the range centered at the median that contains 68% of the counts in Figure 21. (In a Gaussian distribution the 68% confidence interval equals the standard deviation.) In the example of Figure 21, the median m and the 68% confidence interval σ are 0.12 and 1.06 K, respectively. These statistics reflect (1) measurement errors and (2) spatial and temporal variation due to sampling the atmosphere at slightly different locations and times. In order to separate the two effects, we express the measured CHAMP and SAC-C temperatures as follows:

$$T_{\text{champ}}(\vec{x}_1, t_1) = T(\vec{x}_1, t_1) + \varepsilon_{\text{champ}}, \quad (1a)$$

$$T_{\text{sac-c}}(\vec{x}_2, t_2) = T(\vec{x}_2, t_2) + \varepsilon_{\text{sac-c}}, \quad (1b)$$

where $T(\vec{x}, t)$ represents the true temperature at $\vec{x}, t$; $\varepsilon_{\text{champ}}$ and $\varepsilon_{\text{sac-c}}$ are the CHAMP and SAC-C temperature errors including biases. On the basis of equations (1a) and (1b) the mean and standard deviation of the CHAMP and SAC-C temperature difference can be written as

$$m_{\Delta T_{\text{gps}}} = m_{\text{atm}} + \bar{\varepsilon}_{\text{champ}} - \bar{\varepsilon}_{\text{sac-c}}, \quad (2a)$$

$$\sigma_{\Delta T_{\text{gps}}}^2 = \sigma_{\text{atm}}^2 + \sigma_{\text{champ}}^2 + \sigma_{\text{sac-c}}^2 = \sigma_{\text{atm}}^2 + 2\sigma_{\text{leo}}^2, \quad (2b)$$

where

$$\begin{aligned} m_{\text{atm}} &= \text{mean}[T(\vec{x}_1, t_1) - T(\vec{x}_2, t_2)]; \\ \sigma_{\text{atm}}^2 &= \text{var}[T(\vec{x}_1, t_1) - T(\vec{x}_2, t_2)]. \end{aligned} \quad (2c)$$

In the second equation of equation (2b) we have assumed that the measurement errors from CHAMP and SAC-C have the same variance.

[64] To get a handle on the degree of atmospheric variability in space and time as reflected in m_{atm} and σ_{atm} , we turn to the ECMWF analysis. We define

$$\begin{aligned} m_{\text{ecmwf}} &= \text{mean}[T_{\text{ecmwf}}(\vec{x}_1, t_1) - T_{\text{ecmwf}}(\vec{x}_2, t_2)]; \\ \sigma_{\text{ecmwf}}^2 &= \text{var}[T_{\text{ecmwf}}(\vec{x}_1, t_1) - T_{\text{ecmwf}}(\vec{x}_2, t_2)] \end{aligned} \quad (3)$$

and conjecture that

$$\sigma_{\text{atm}}^2 > \sigma_{\text{ecmwf}}^2, \quad (4)$$

$$m_{\text{atm}} \approx m_{\text{ecmwf}}. \quad (5)$$

Equation (4) implies that the model underestimates the true variability of the atmosphere. Equation (5) implies that even if the analysis is biased, there is little ground to believe that temperature differences between nearby grids are biased. Using equations (2a), (2b), (4), and (5), we can write

$$\sigma_{\text{leo}}^2 = \frac{1}{2} (\sigma_{\Delta T_{\text{gps}}}^2 - \sigma_{\text{atm}}^2) < \frac{1}{2} (\sigma_{\Delta T_{\text{gps}}}^2 - \sigma_{\text{ecmwf}}^2), \quad (6a)$$

$$\bar{\varepsilon}_{\text{champ}} - \bar{\varepsilon}_{\text{sac-c}} = m_{\Delta T_{\text{gps}}} - m_{\text{atm}} \approx m_{\Delta T_{\text{gps}}} - m_{\text{ecmwf}}. \quad (6b)$$

[65] Since σ_{atm}^2 is not directly available to us but σ_{ecmwf}^2 can be computed from the analysis, equation (6a) sets an upper bound on the error of GPS-LEO occultation measurements by CHAMP and SAC-C.

[66] Figure 22 shows the square root of the right-hand side (RHS) of the inequality of equation (6a) as a function of height and cutoff distance $\bar{d}$ ($\bar{d} = 100, 200$ and 300 km). In computing the RHS of equation (6a) for a given height $\bar{h}$ and distance cutoff $\bar{d}$, all points in S satisfying $|h - \bar{h}| < 2.5$ km and $d < \bar{d}$ are chosen and the median and 68% confidence interval are computed. As $\bar{d}$ approaches 0, the standard deviation curve becomes smaller and approaches the uncertainty in the GPS occultation temperature measurements. This is expected on the basis of equation (6a) since σ_{ecmwf}^2 would reach 0. The 68% confidence interval curve associated with 100 km shows a $\sigma_{\text{leo}} < 0.6$ K between 5 and 15 km. The error grows to ~2 K at 25 km and to ~6 K at 35 km altitudes. The curve does not extend to the surface because we exclude regions in the troposphere where $T > 250$ K to avoid the dry/moist term ambiguity.

[67] Similarly, we can infer the mean difference between CHAMP and SAC-C by computing the RHS of equation (6b). The results are shown in Figure 22 for similar cutoff distances as before. Mean differences are less than 0.1 K at all heights below 18 km, reflecting the precision of

averaged GPS profiles. Even smaller differences may be seen when averaging over larger data sets (212 or less were used here) used for climate variability studies. This result tends to support one of the more dramatic claims made for GPS sounding in early analytical studies [Kursinski *et al.*, 1997] that its inherently low bias and high precision can provide regional temperature averages accurate to better than 0.1 K (or 1 part in 3000), enabling the monitoring of very small changes in climate with a technique that is at once efficient, global, and all-weather.

[68] It is important to keep in mind possible systematic biases that can affect CHAMP and SAC-C in a similar way. For instance, the discussion in section 4.3 shows that a bias of 2K in the initialization temperature (used to initialize the hydrostatic integral at 40 km) would cause a bias of 0.03 K in the retrieved temperature at 15 km. Therefore, if GPS RO climatological records of temperature are to be compared over extended periods (years to decades) for the purpose of monitoring climate trends, it is important to understand and control the initialization temperature biases (which themselves can change over time) such that they do not appear as trends in the GPS climatological records. Similarly, ionospheric residual effects can also be common to both CHAMP and SAC-C (although not exactly the same since the coincident occultations are often seen from different directions). Understanding and controlling these ionospheric residuals to a sufficiently small level is very important in order not to alias solar fluctuations affecting the ionosphere into atmospheric climate signals. While these are issues that need to be addressed in some depth to understand their long-term implications, this study provides a first step toward establishing the level of consistency between different GPS measurements as a means of estimating their precision.

8. Summary and Conclusion

[69] By June 2003, nearly half a million CHAMP and SAC-C occultations have been collected and processed to yield high-quality temperature profiles in the upper troposphere and lower stratosphere. These occultations are obtained during a time when the GPS L2 signal is encrypted. Comparisons to weather analysis indicates that the quality of retrievals is not degraded compared to data collected from the GPS/MET experiment during periods when the encryption of the GPS signal was temporarily disabled supporting previous findings [Wickert *et al.*, 2001; Marquardt *et al.*, 2003]. This is made possible by a new generation of GPS flight receivers employing advanced “semicodeless” techniques and is vital to establish GPS sounding as a powerful new tool for climate and weather research.

[70] The two missions produce up to 425 daily global occultations a number that approaches the expected level of 500. Future software uploads and the enabling of GPS occultations on GRACE will bring the number of daily soundings to over 1200, surpassing the total of semidaily radiosonde profiles.

[71] The enhanced tracking loop in the lower troposphere currently enables the signal to reach to the lowest 0.5 km for ~60% of the time globally (35% in the tropics and 85% at latitudes >60 degrees). This is a considerable improvement over the performance of GPS/MET and this number is

expected to reach 90–95% when the “open loop” tracking is implemented.

[72] Collection of precise GPS soundings concurrently from two platforms has enabled for the first time direct inter-comparisons of profiles. Our analysis shows that temperature profiles collected from CHAMP and SAC-C are consistent to 0.1 K in the mean and 0.86 K in standard deviation between 5 and 15 km. If the errors in CHAMP and SAC-C are assumed to be uncorrelated, this implies that individual profiles are precise to <0.6 K between 5 and 15 km.

[73] A precision of 0.1 K in the mean makes GPS sounding ideally suited to monitor weak climate trends. Assuming a 0.2 K/decade temperature increase (the midpoint of IPCC estimates for human-driven warming [Houghton *et al.*, 1990, p. 248]), the GPS measurement would be sensitive to this change within about 5 years. This sensitivity does not answer the bigger and more difficult question of separating natural and anthropogenic forcing, but it is a necessary step for addressing it.

[74] Several challenges remain including the negative N-bias caused by super refraction, the nonperfect cancellation of the clock appearing as a 1 Hz signal, improved L2 tracking in the lower troposphere, improved understanding of the effect of horizontal gradient on the retrievals, the nonperfect cancellation of the ionosphere and its impact on long term atmospheric climate records obtained with GPS occultations, and the optimal method(s) of assimilating this type of data into numerical weather prediction models.

[75] With the many planned missions to collect GPS soundings in the next decade, including COSMIC, ACE+, EQUARS, NPOESS, and other flights of opportunities, the density of coverage will become sufficient for a wide range of climate and weather applications, including establishing detailed baseline climatologies; conducting studies of climate processes, such as stratosphere-troposphere exchange and tropical convection; testing climate models; and detecting climatic change. GPS soundings have the potential to constitute a vital input both to general circulation models and to numerical weather models and to trigger improvements in a range of environmental predictions, from tomorrow's weather forecast to seasonal climate variations to long-term climatic change.

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